

ECEN 474/704: Lab 07

Frequency response of Inverting Amplifier (Simulating 4 types of inverting amplifiers)

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Due date: 2025-11-14

I. DISCUSSION OF THE LAB AND RESULTS

Include a short write-up (a paragraph or two) summarizing the lab and results. Compare the final designs with your pre-lab designs (as applicable for the lab), and make note of any major changes in your designs and why these changes were made. Regarding the simulation results, briefly describe any trends that are shown from the data or differences between pre- and post-layout simulation results, and attempt to provide an explanation for the trends and/or differences. Additionally, consider some of the following questions: were the results as you expected? Why or why not? What, if any, design trade-offs are demonstrated by the work in the given lab? If you were unable to meet any specifications given in the lab, what do you believe was the primary cause and what could be some potential solutions for the future?

The above questions are not an exhaustive list of what could be discussed and may not be relevant for all labs, but rather they serve as a starting point for you to demonstrate your understanding of the lab material.

In this lab for all 4 inverting amplifiers, the Current-Mirror Load, Digital CMOS, Self-Biased PMOS, Self-Biased CMOS, DC analysis was done verifying if all the transistors are brought to saturation within the voltage range 0.9 to -0.9 (VDD to VSS) and if done, the operating point was found and is fixed in the test bench to perform the AC Analysis and then by trial and error method the widths of both p-devices and n-devices are continuously changed and varied to arrive at the desired specifications. Once the widths and the lengths are decided, then the layout for the schematic is done and then checked for DRC and LVS errors. Once the DRC and LVS is cleared, then pex extraction is done and the pre and the post layout simulation is compared

1. Inverter with Current-Mirror Load:

In this circuit simulation the gain and gmw was kind of trade-off between each other, when gmw was tied to be adjusted to meet the specification the gain decreased from 30db to 29 db. The miller capacitance was also decreased to achieve the desired GBW

2. Digital CMOS:

After so much trial and error, could not bring the gain above 29 db. The miller capacitance was also decreased to achieve the desired GBW

3. Self-Biased PMOS:

In this circuit simulation, the gbw is same as the first pole frequency which is the frequency at -3db graph

4. Self-Biased CMOS:

In this circuit simulation, the gbw is same as the first pole frequency which is the frequency at -3db graph

II. SUMMARY OF RESULTS

Summarize the simulation results from the lab in one or two tables, like what is shown below.

Table 1
SUMMARY OF RESULTS FOR CURRENT MIRROR LOAD(IBIAS CURRENT)

Parameter	WIDTH OF P-DEVICE (length-270nm)dum my-nf=4	WIDTH OF N DEVICE (length-270nm)- dummy-nf=4	NUMBER OF FINGERS	A0(GA IN)	MILLER CAP	DC OPERATING POINT(m V)	GBW	UGF
PRE-LAYOUT	10u	20u	20	30.1846	0.682pF	-322.797	2.0723	89.7097
POST-LAYOUT	10u	20u	40	29.727	0.682pF	-323.106	2.0723	90.299

Table 2
SUMMARY OF RESULTS FOR DIGITAL CMOS INVERTER

Parameter	WIDTH OF P-DEVICE(le ngth-270nm) dummy- nf=4	WIDTH OF N DEVICE (length-270nm) dummy- nf=2	NUMBER OF FINGERS	A0(GA IN)	MILLER CAP	DC OPERATING POINT(m V)	GBW	UGF
PRE-LAYOUT	28u	5u	56	28.246	0.661pF	67.318	2.087	90.9964
POST-LAYOUT	28u	5u	10	28.215	0.661pF	29.72911	2.0864	90.9674

Table 3
SUMMARY OF RESULTS FOR SELF-BIASED PMOS

Parameter	WIDTH OF P-DEVICE- (length-270nm)Dum my-nf-10	WIDTH OF N DEVICE (length-270nm)	NUMBER OF FINGERS	A0(GA IN)	MILLER CAP	DC OPERATING POINT(m V)	GBW	UGF
PRE-LAYOUT	20u	-	40	0	0.24 pF	312.0404m V	2.121	112.82

Parameter	WIDTH OF P-DEVICE-(length-270nm)Dummy-nf-10	WIDTH OF N DEVICE (length-270nm)	NUMBER OF FINGERS	A0(GA IN)	MILLER CAP	DC OPERATING POINT(m V)	GBW	UGF
POST-LAYOUT	10u	-	20	0	0.24pF	299.1914m V	1.7188	111.119

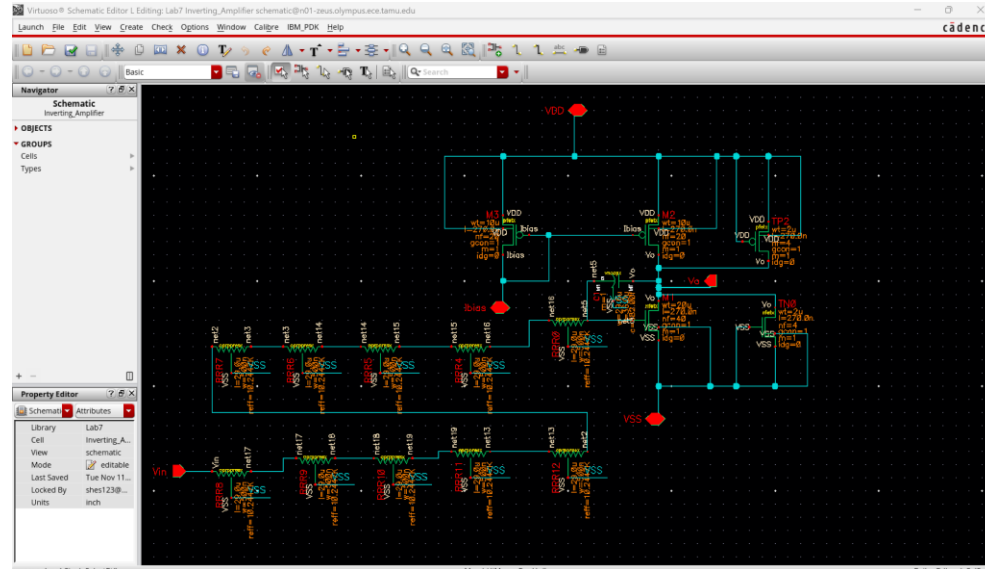
Table 4
SUMMARY OF RESULTS FOR SELF BIASED CMOS

Parameter	WIDTH OF P-DEVICE(length-180nm)Dummy nf=8	WIDTH OF N DEVICE(length-180nm) Dummy nf=6	NUMBER OF FINGERS	A0(GA IN)	MILLER CAP	DC OPERATING POINT(m V)	GBW	UGF
PRE-LAYOUT	12 u	6 u	24	0	0.24	-71.23614	2.749	121.154
POST-LAYOUT	12 u	6 u	12	0	0.24	-128.9907	2.749	121.154

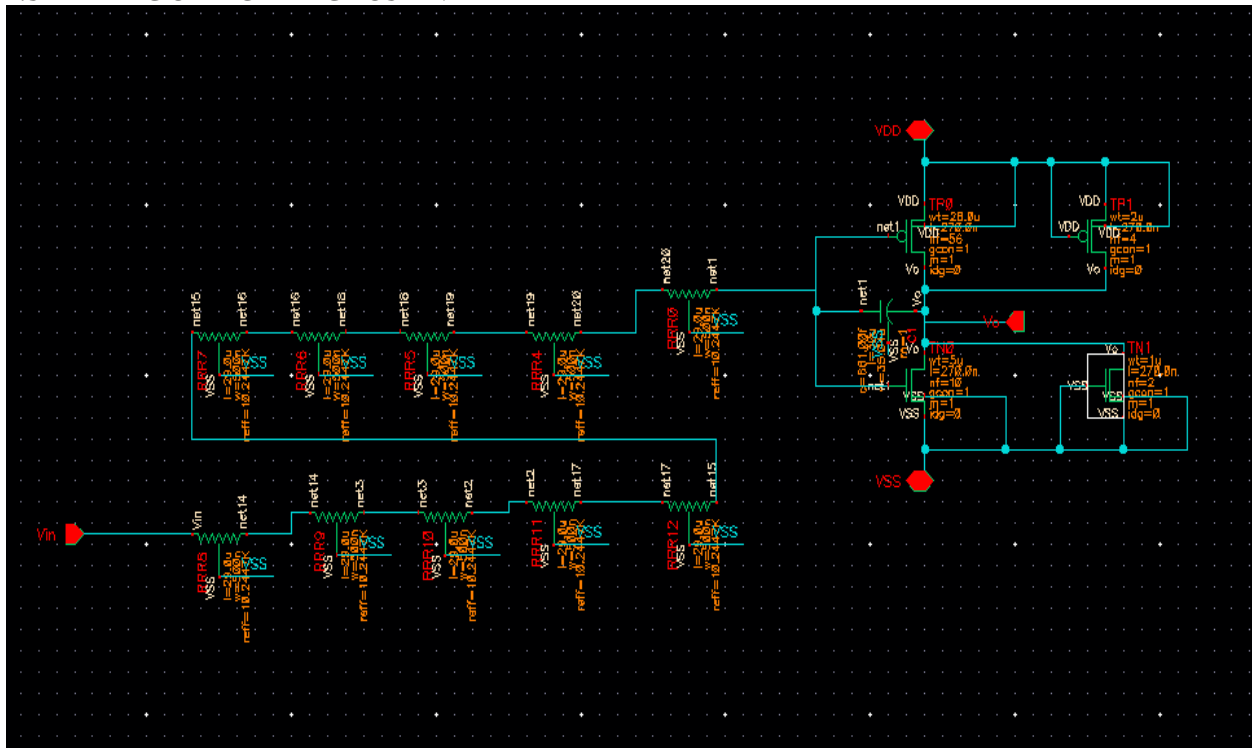
III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

Schematics of the sub-circuit(s) being designed in the lab should be included here. In addition, include images of the test bench schematics. Important signals in the test bench schematic should be labeled to provide clear correspondence between the signals in the schematics and the signals that are plotted in the simulation results shown later in the report.

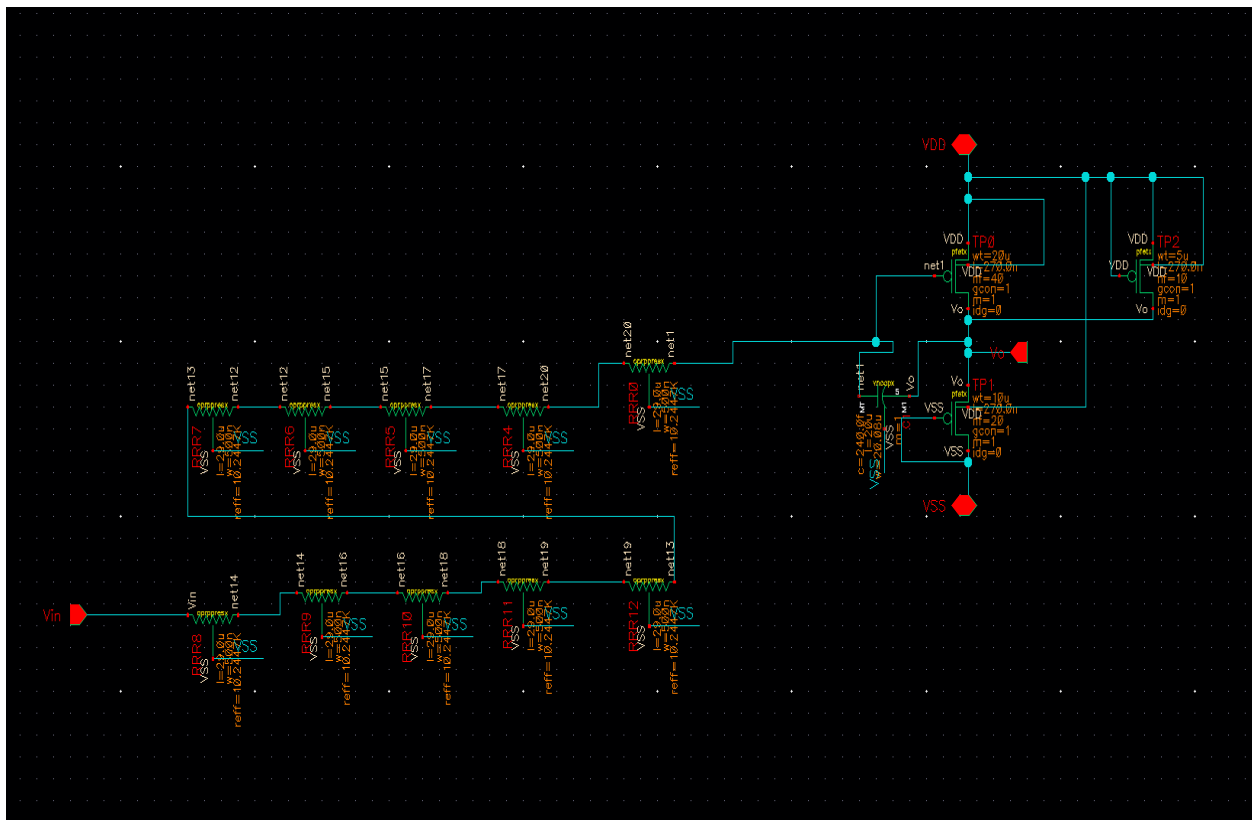
1.SCHEMATIC OF CURRENT MIRROR LOAD INVERTER



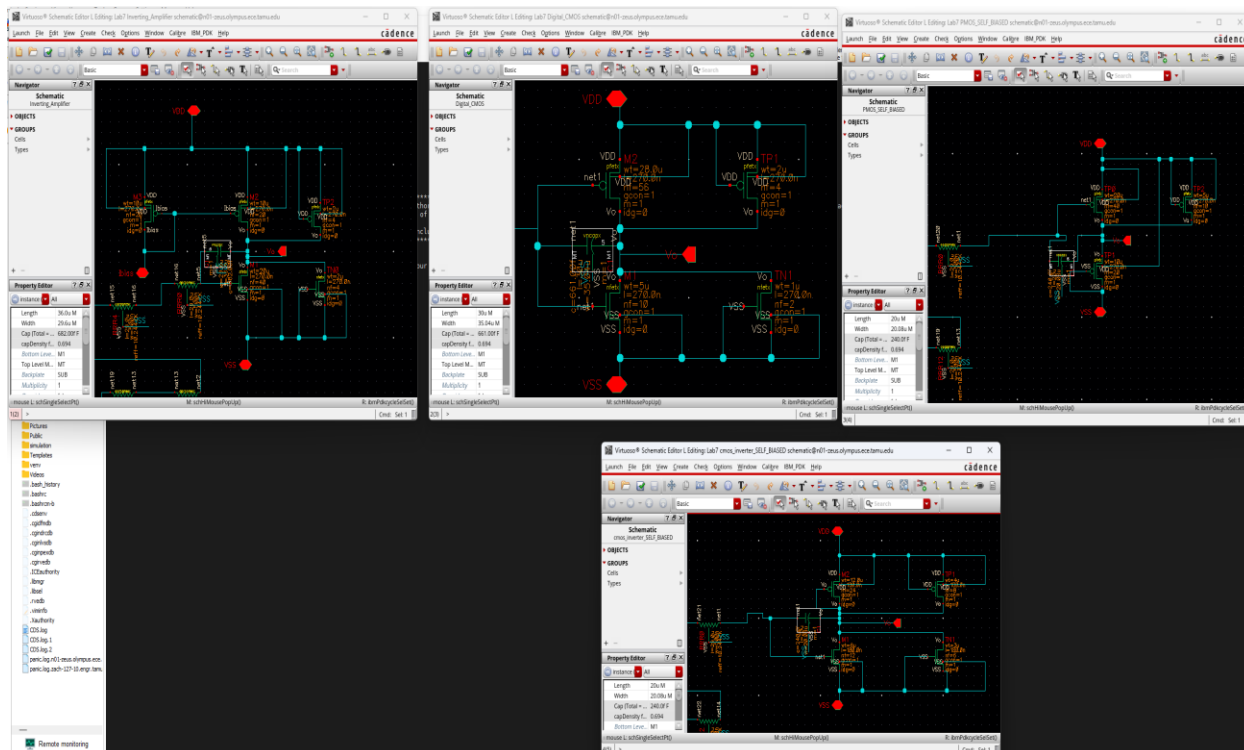
2.SHEMATIC OF DIGITAL CMOS INVERTER



3.SHEMATIC OF SELF-BIASED PMOS INVERTER

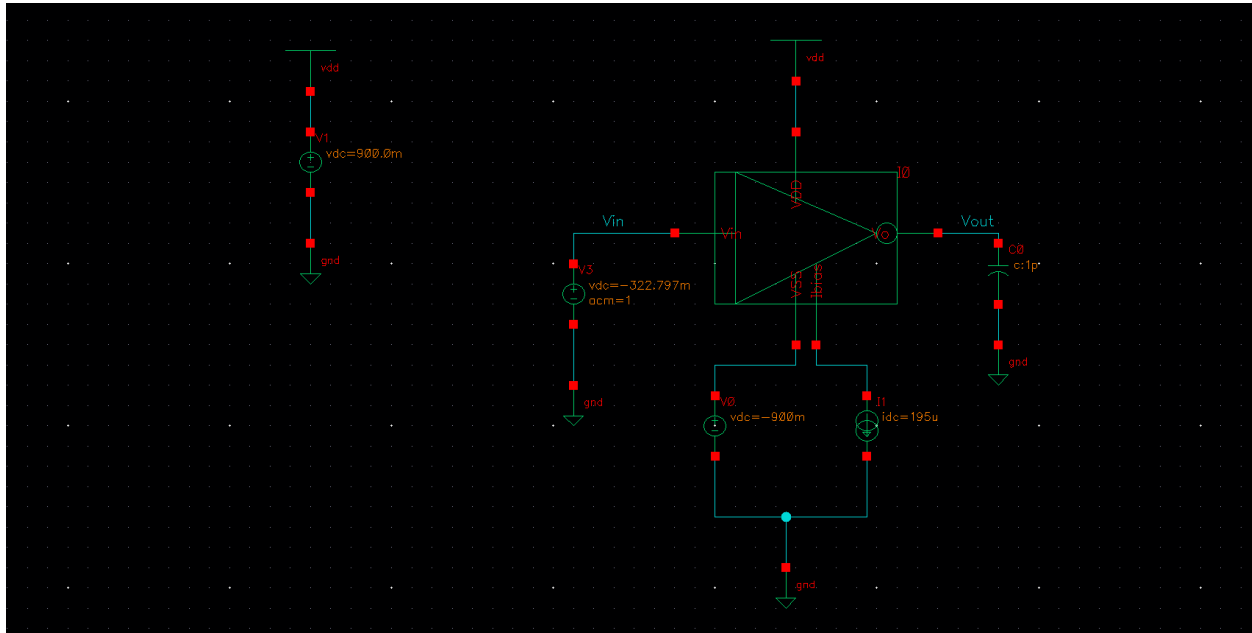


4:SCHEMATIC OF SELF-BIASED CMOS

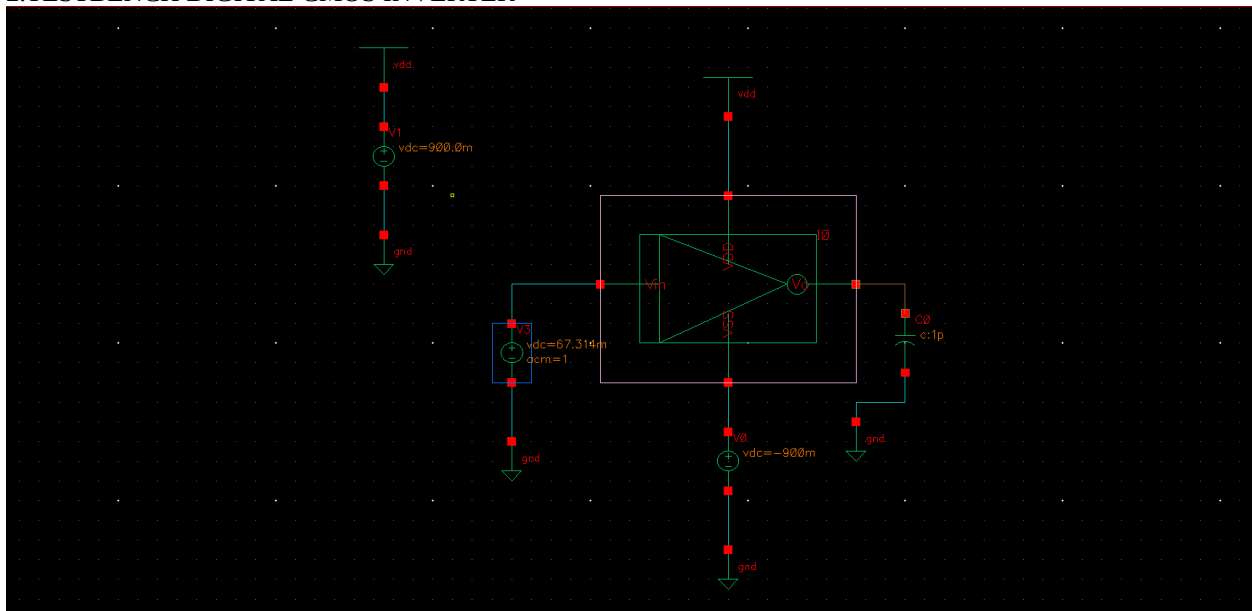


TEST BENMCHES:

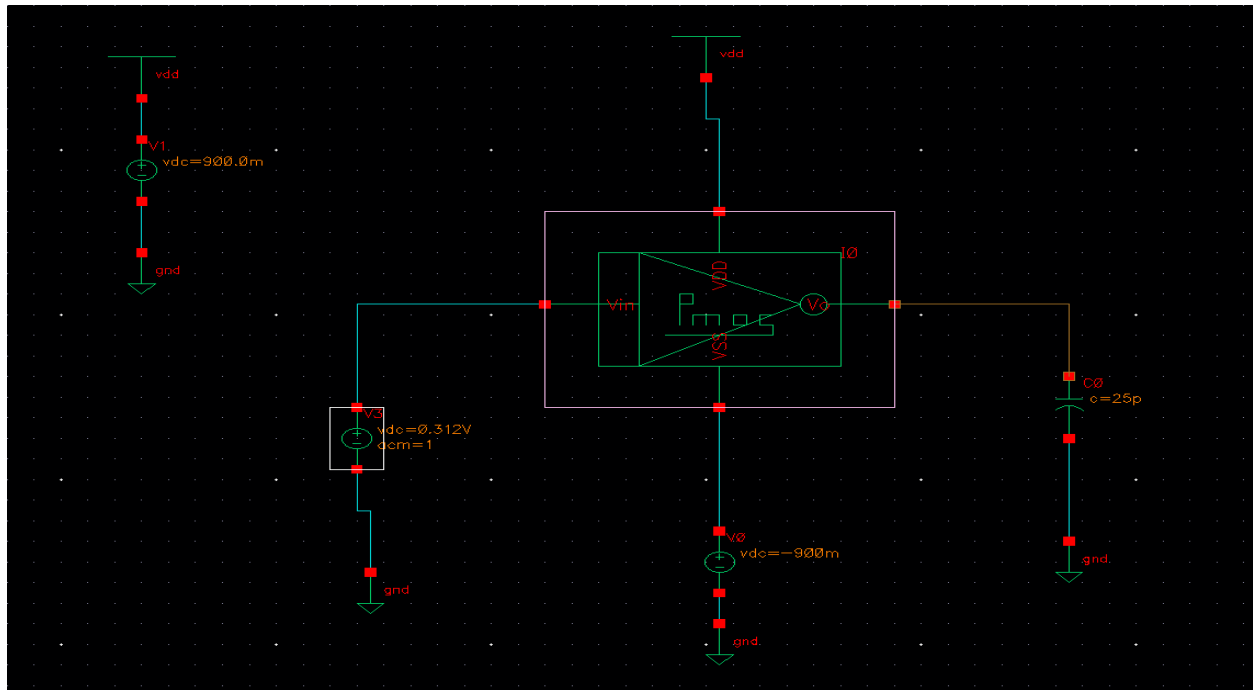
1.TESTBENCH OF CURRENT-MIRROR LOAD



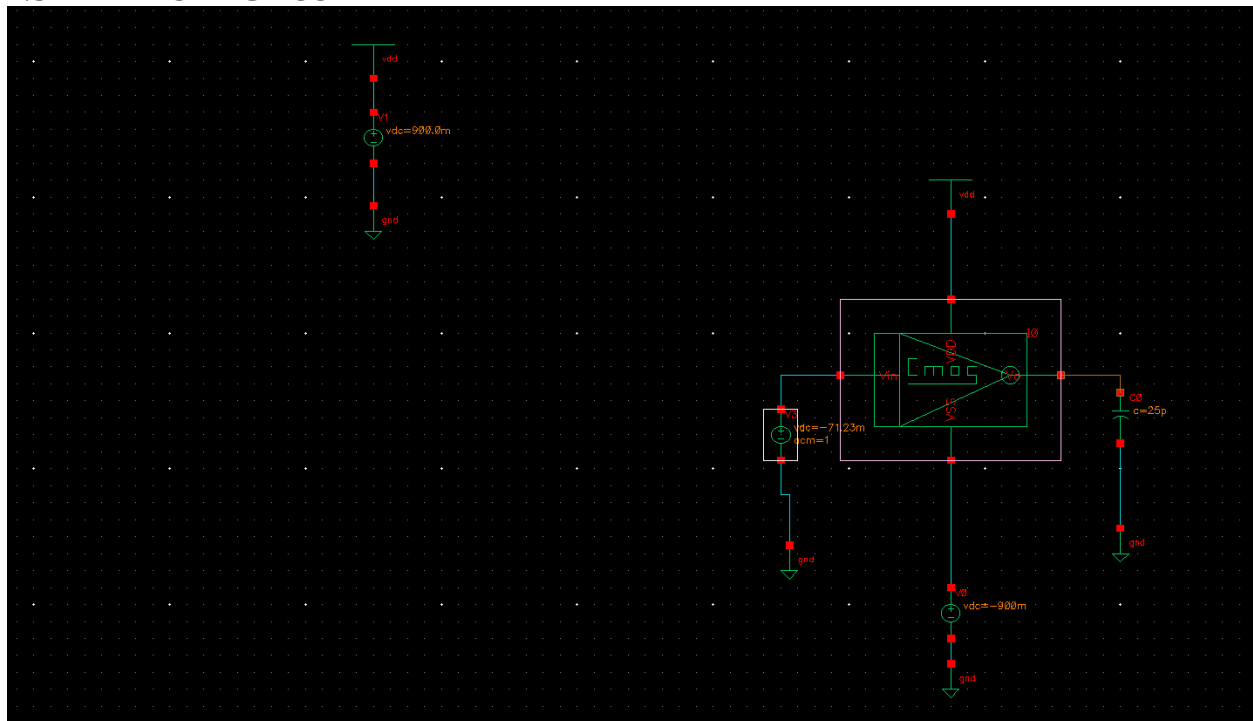
2.TESTBENCH DIGITAL CMOS INVERTER



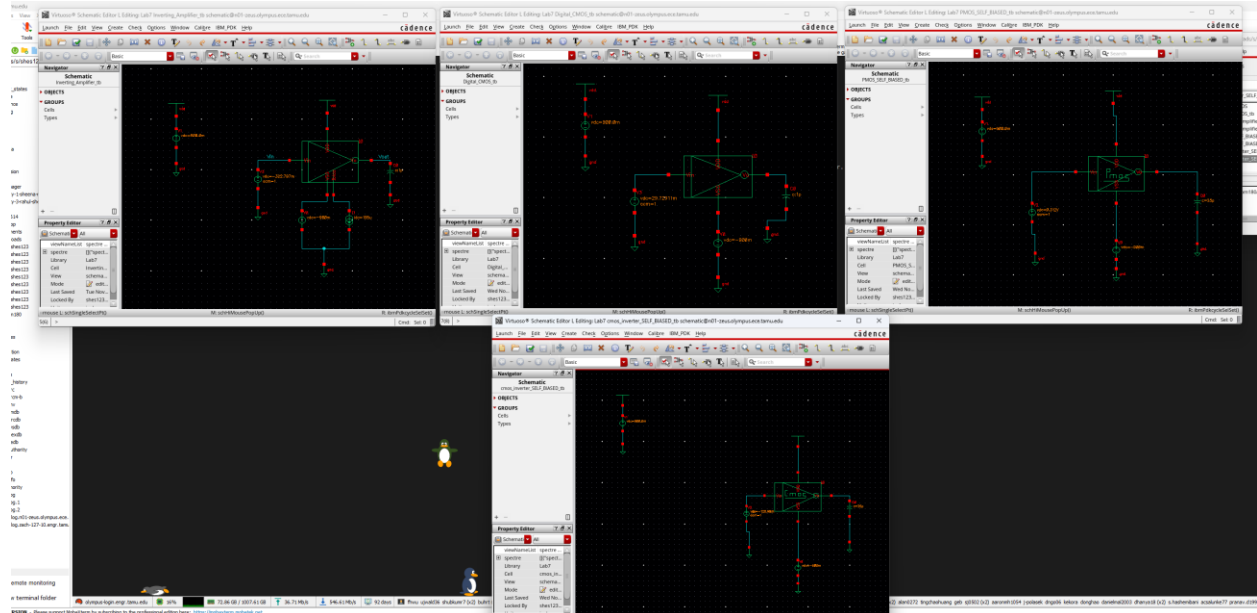
3.TESTBENCH OF SELF-BIASED PMOS



4.SELF-BIASED CMOS



Overall view:

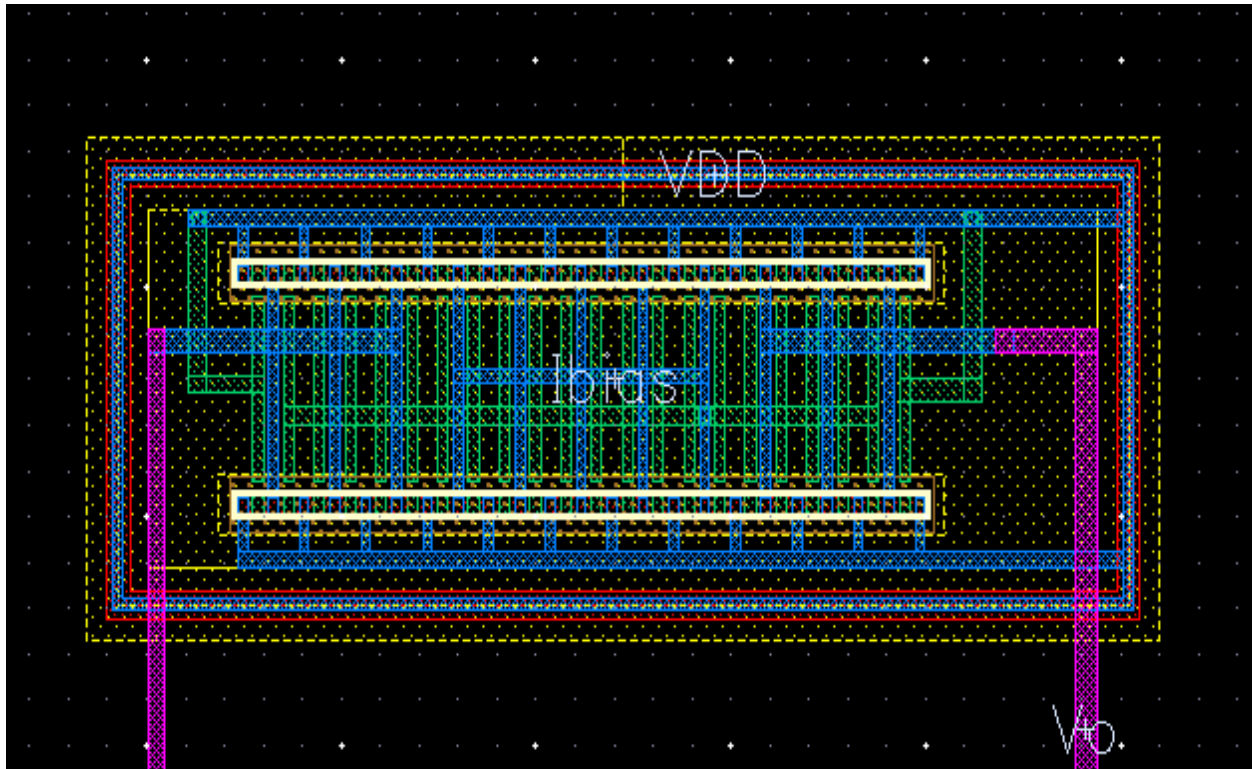


IV. LAYOUT(S) AND FLOOR PLAN(S)

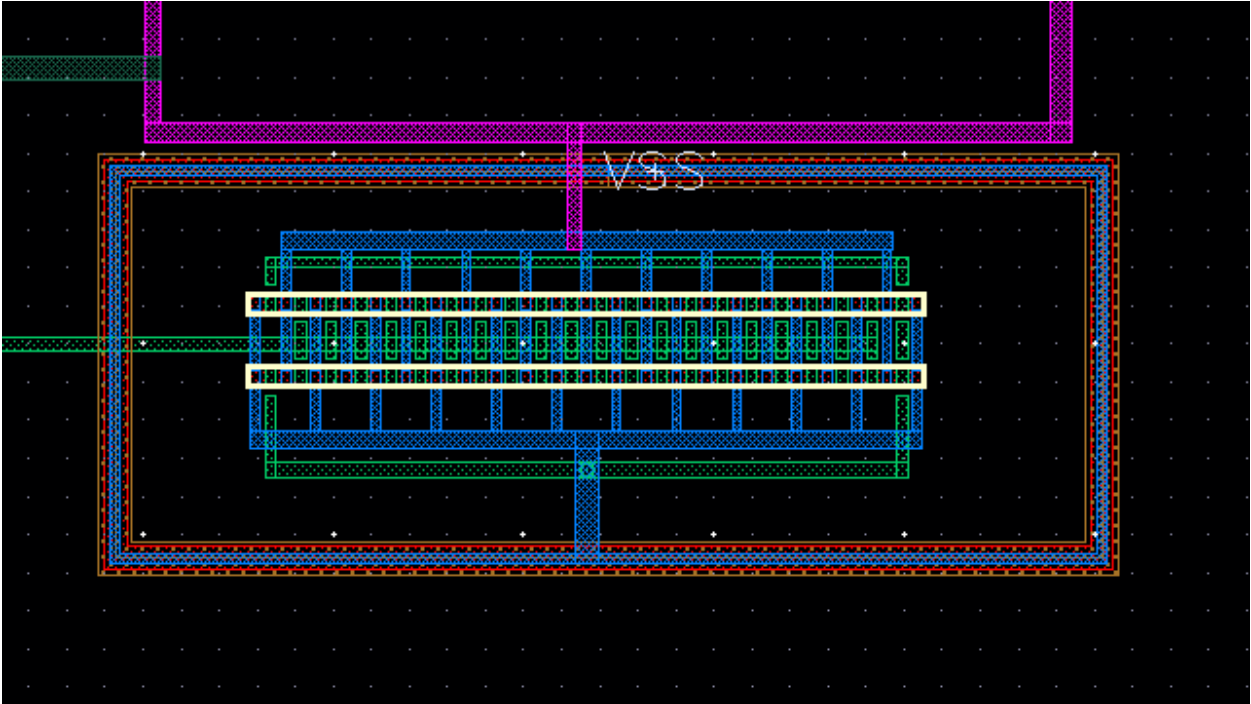
Include diagrams of the floor plans used to lay out major circuit components. Typically, this will just include transistor floor plans (routing does not need to be included in the diagrams). A table can be used to show the layout arrangement for the transistors or the floor plan can be drawn in another software tool. Clearly label the devices in the floor plan and make sure there is clear correspondence with the devices in the schematics, and include dummy devices in the floor plan diagram (dummy devices are often denoted with an asterisk, *).

This section will also include images of the layout(s) of each circuit constructed for the lab.

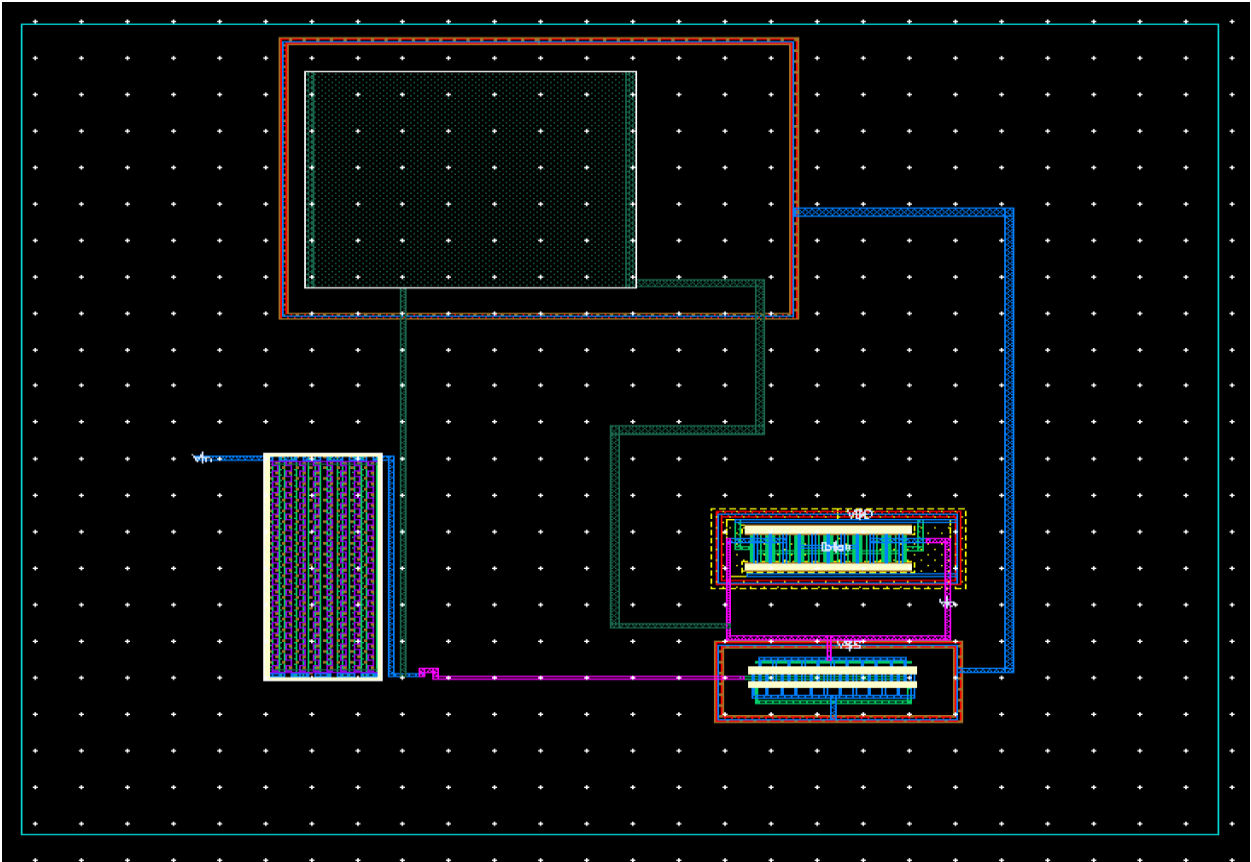
1. LAYOUT OF CURRENT MIRROR LOAD INVERTING AMPLIFIER CLOSER LOOK OF P-DEVICES



CLOSER LOOK OF N DEVICES



OVERALL LAYOUT

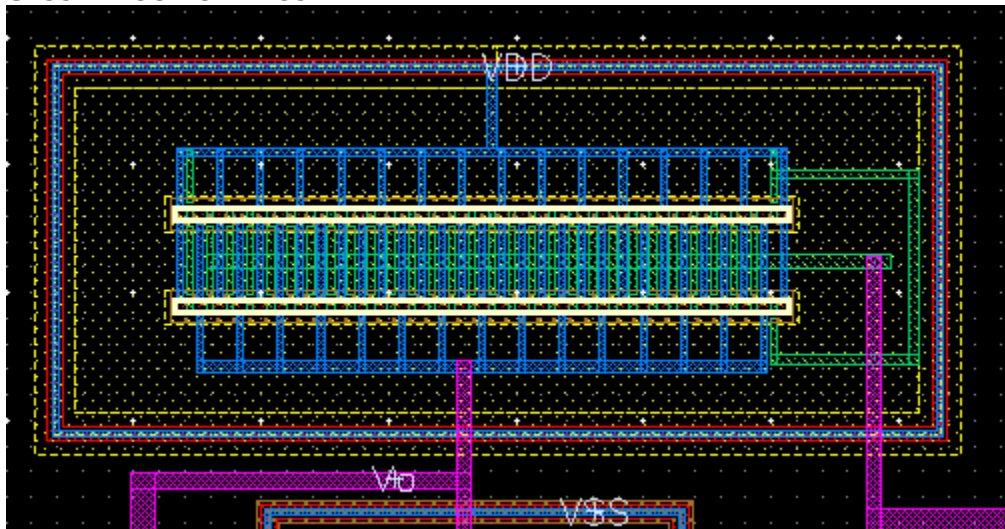


FLOOR PLAN:

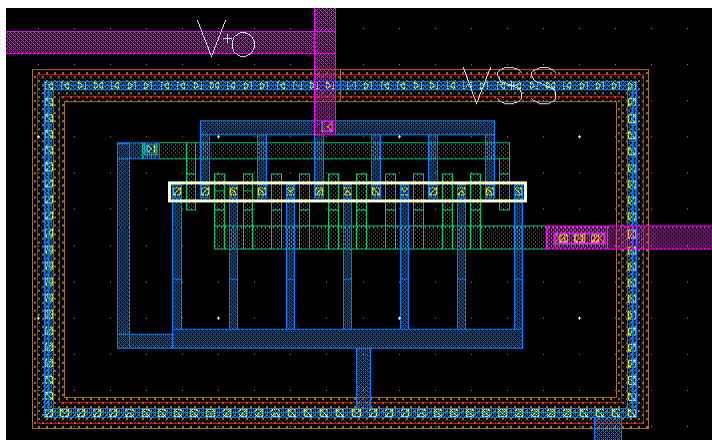
FLOOR PLAN 1

- *M2M2M2M2M2M3M3M3M3M3M3M3M3M2M2M2M2M2*
- *M2M2M2M2M2M3M3M3M3M3M3M3M3M2M2M2M2M2*
- *M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1*
- *M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1M1*

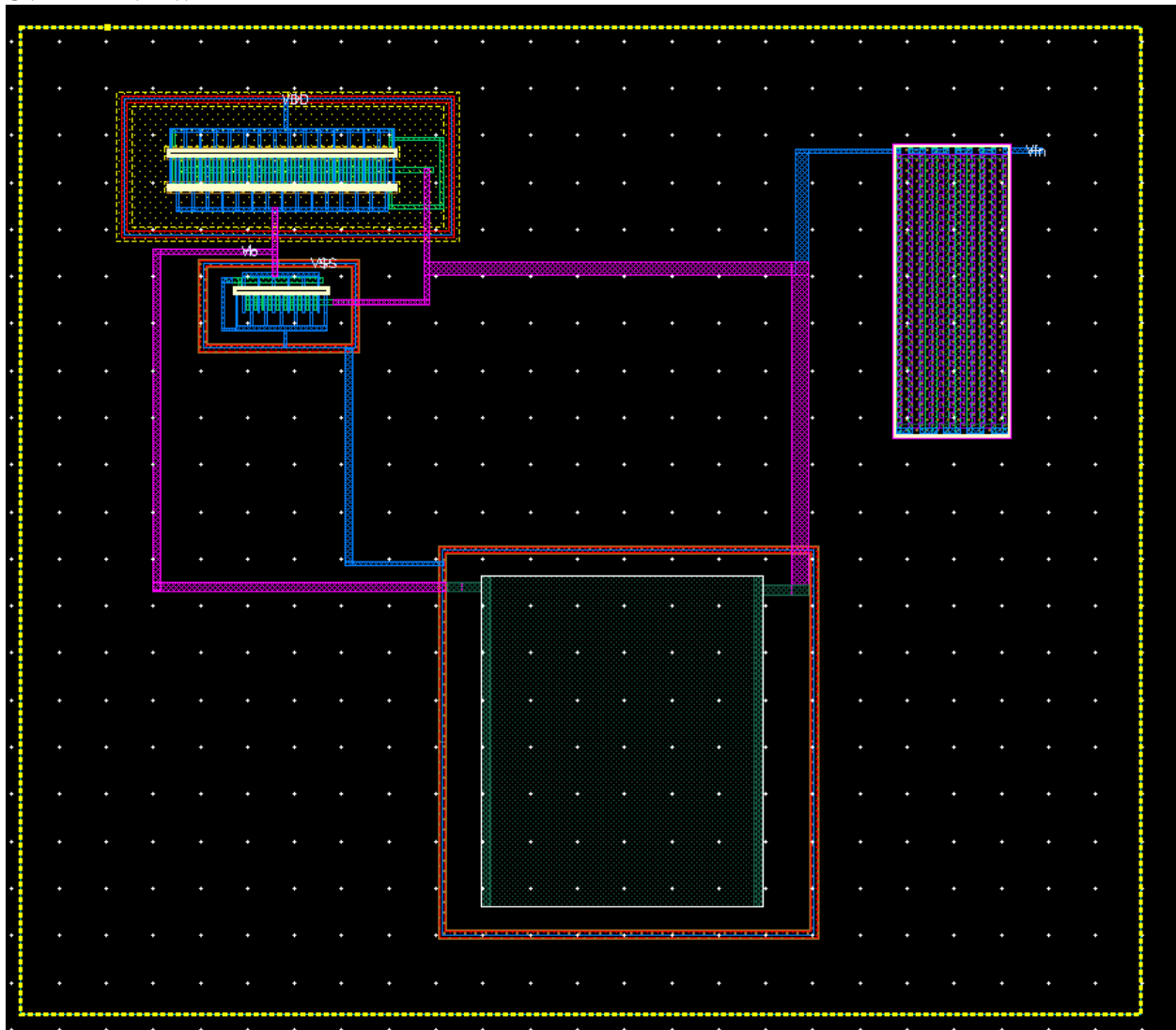
2.LAYOUT OF DIGITAL CMOS INVERTER CLOSER LOOK OF PMOS



CLOSER LOOK OF N DEVICES



OVERALL VIEW

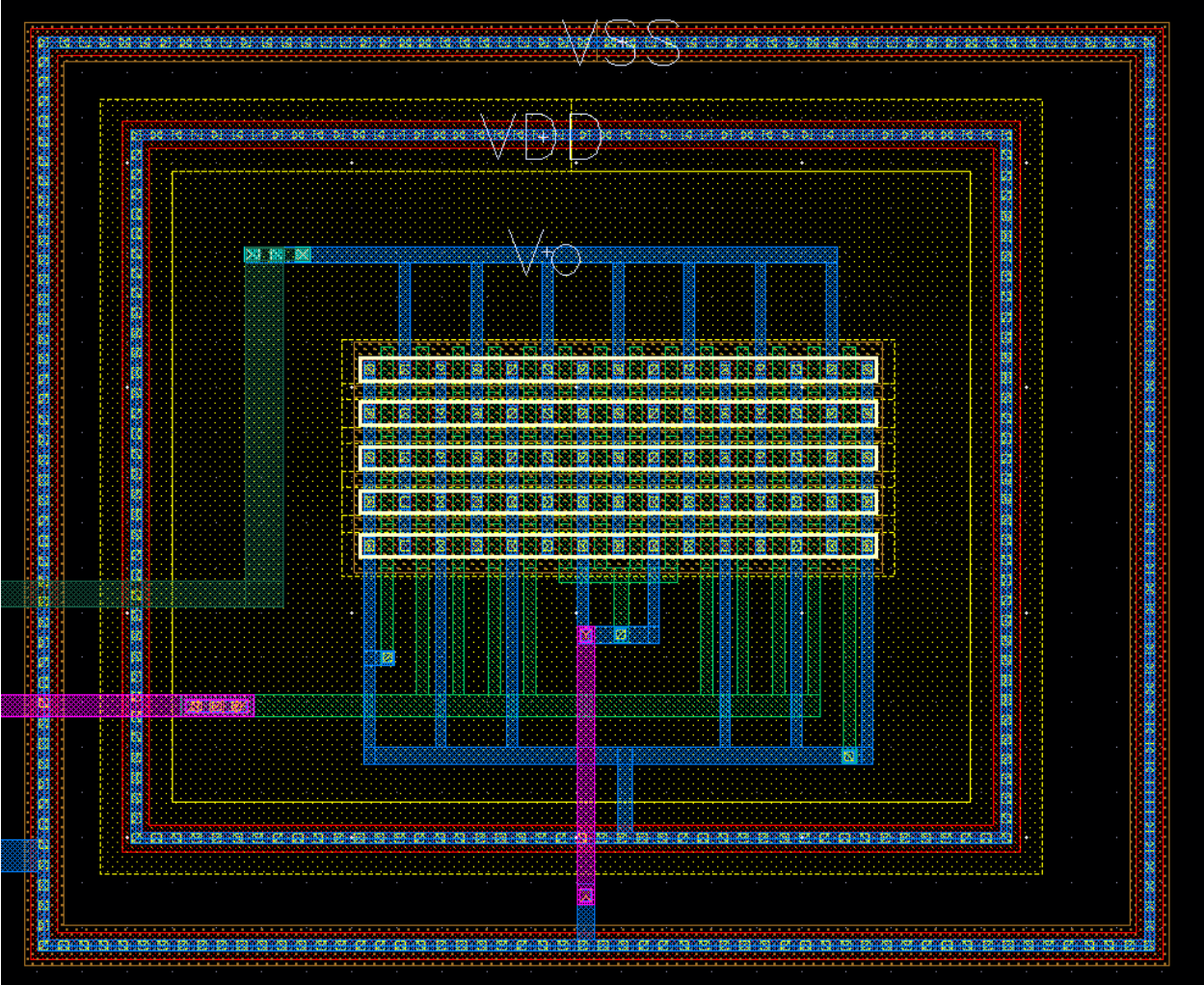


FLOOR PLAN:

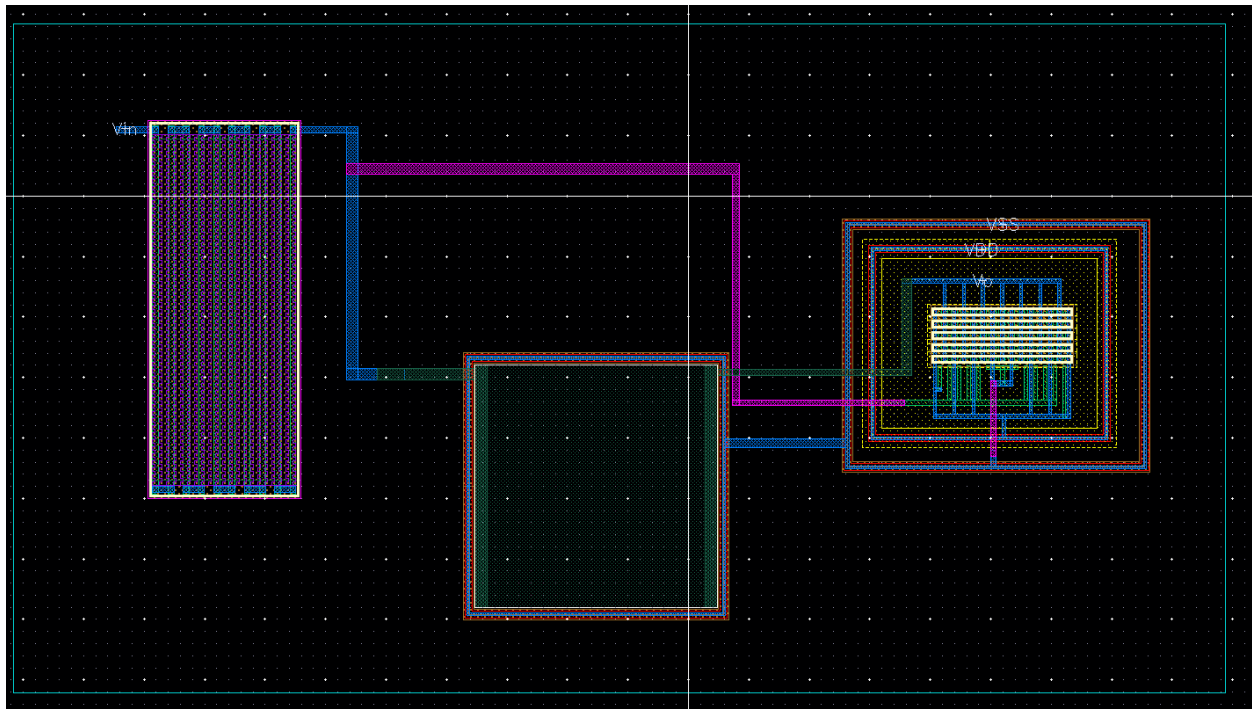
FLOOR PLAN 2

- *M2*
- *M2*
- *M1*

3.LAYOUT OF SELF BIASED PMOS ONLY INVERTER CLOSER LOOK OF P-DEVICES



OVERALL VIEW



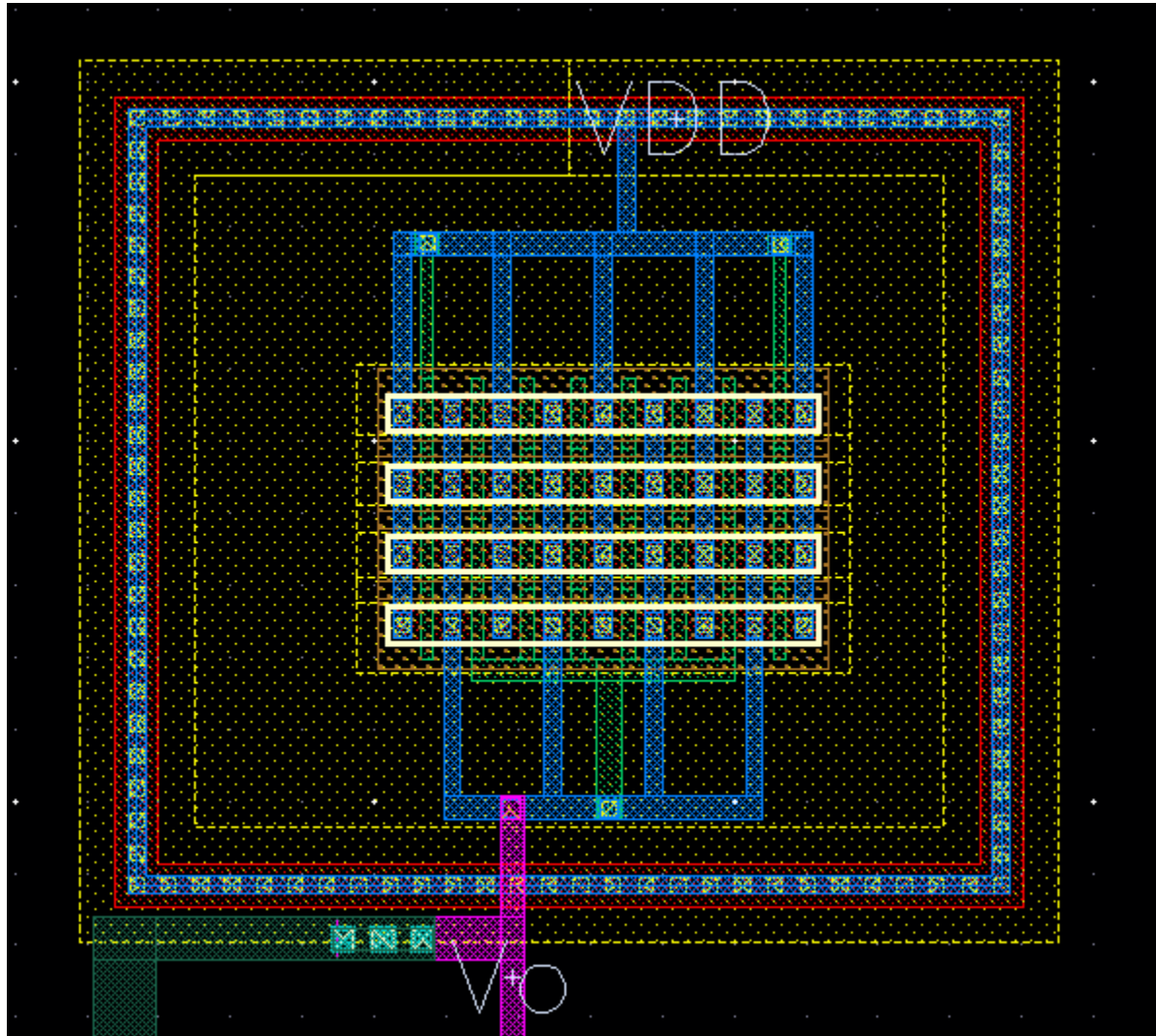
FLOOR PLAN:

FLOOR PLAN 3

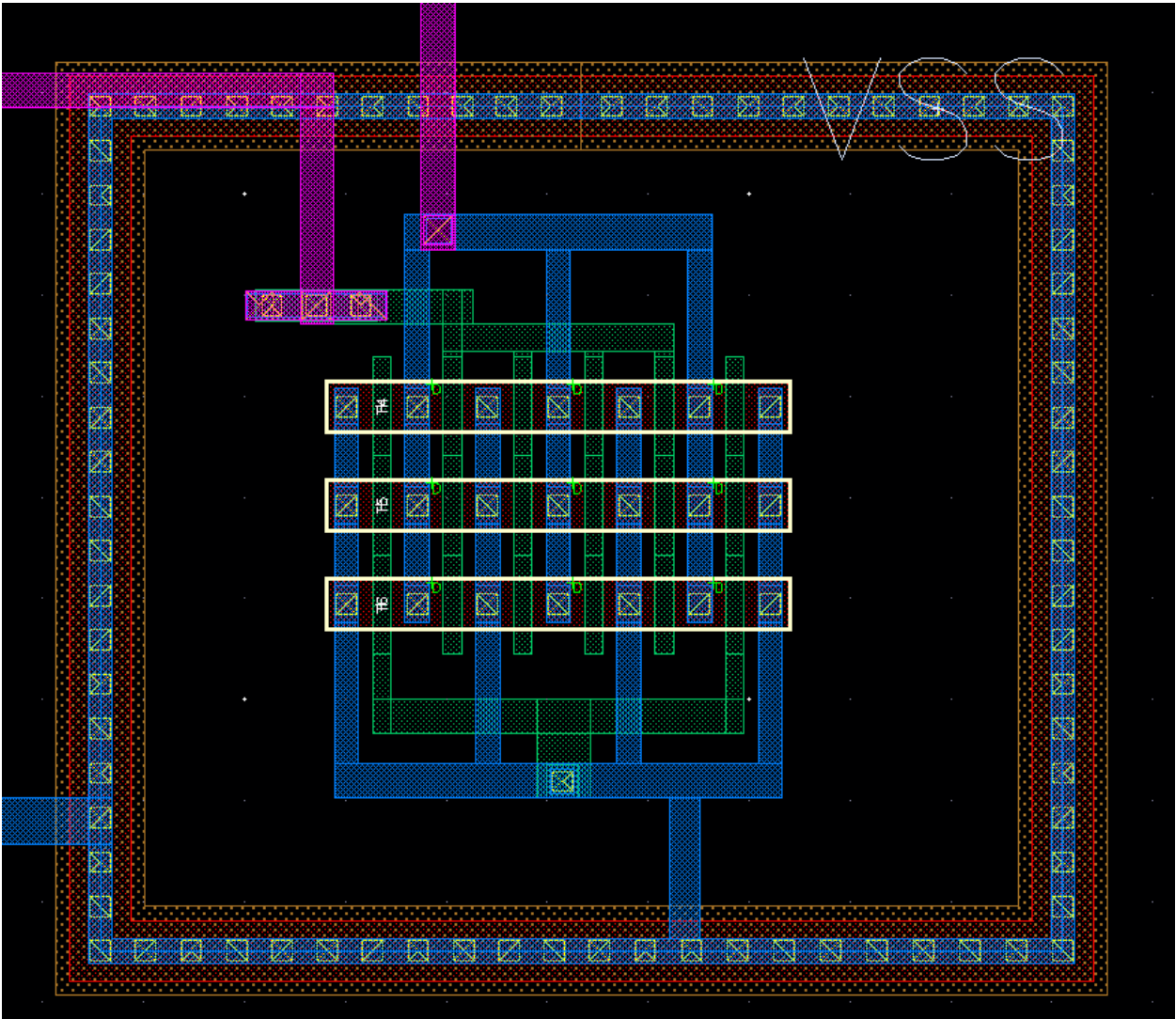
- *M1M1M1M1M2M2M2M2M1M1M1M1*
- *M1M1M1M1M2M2M2M2M1M1M1M1*
- *M1M1M1M1M2M2M2M2M1M1M1M1*
- *M1M1M1M1M2M2M2M2M1M1M1M1*
- *M1M1M1M1M2M2M2M2M1M1M1M1*

4, SELF-BIASED CMOS INVERTER

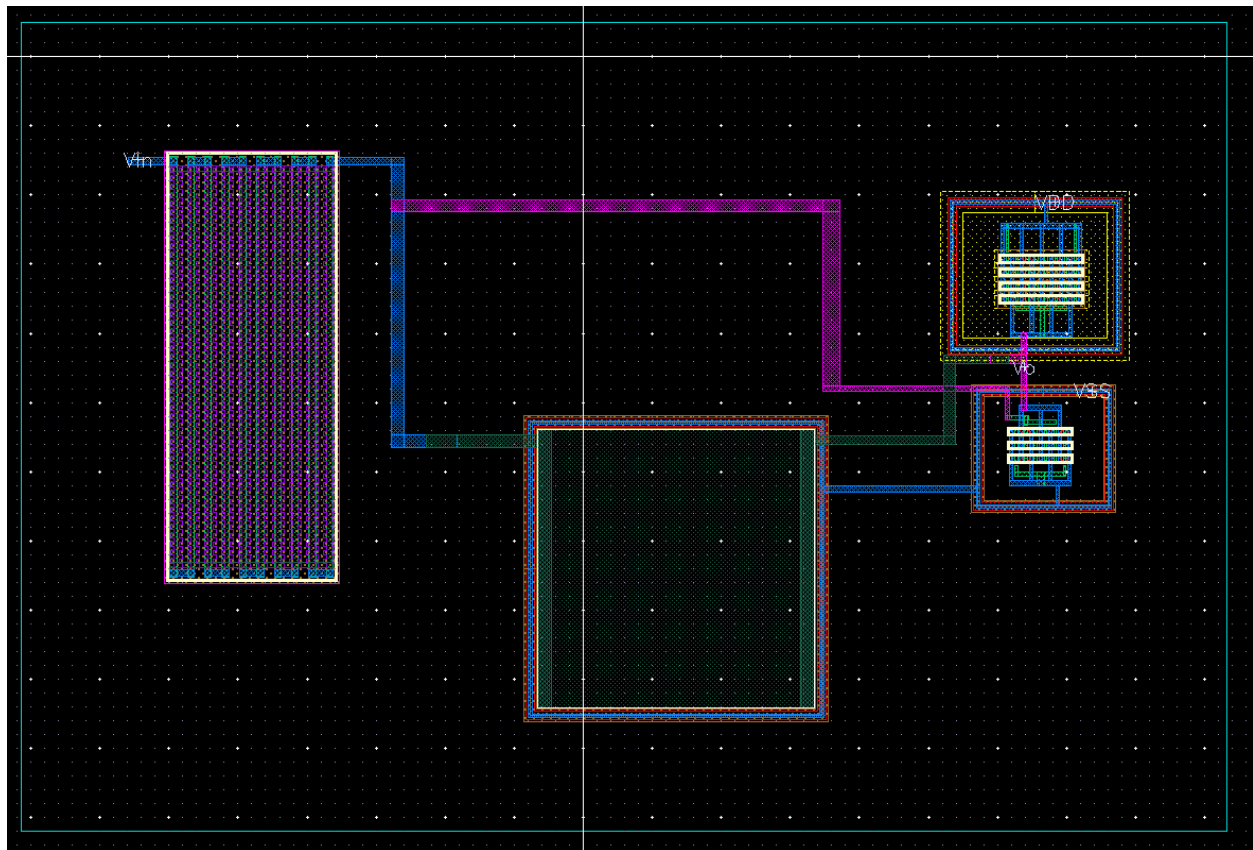
CLOSER LOOK OF P-DEVICES



CLOSER LOOK OF N-DEVICES



OVERALL VIEW

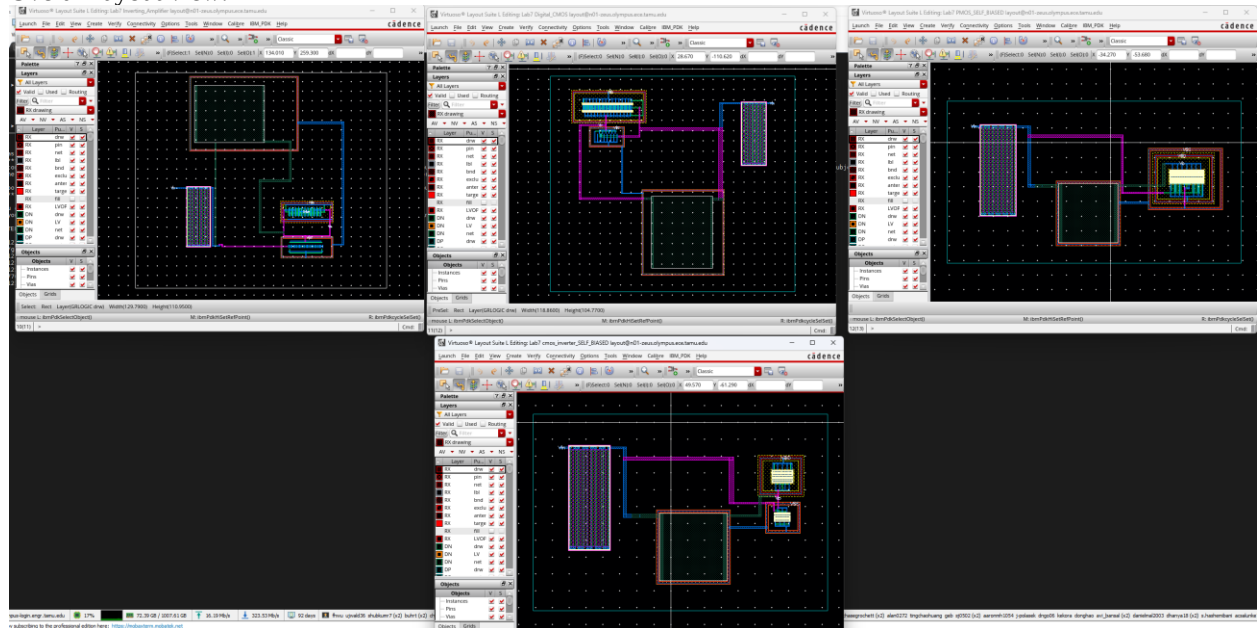


FLOOR PLAN:

FLOOR PLAN:4

- *M2M2M2M2M2M2*
- *M2M2M2M2M2M2*
- *M2M2M2M2M2M2*
- *M2M2M2M2M2M2*
- *M1M1M1M1*
- *M1M1M1M1*

Overall layout view:

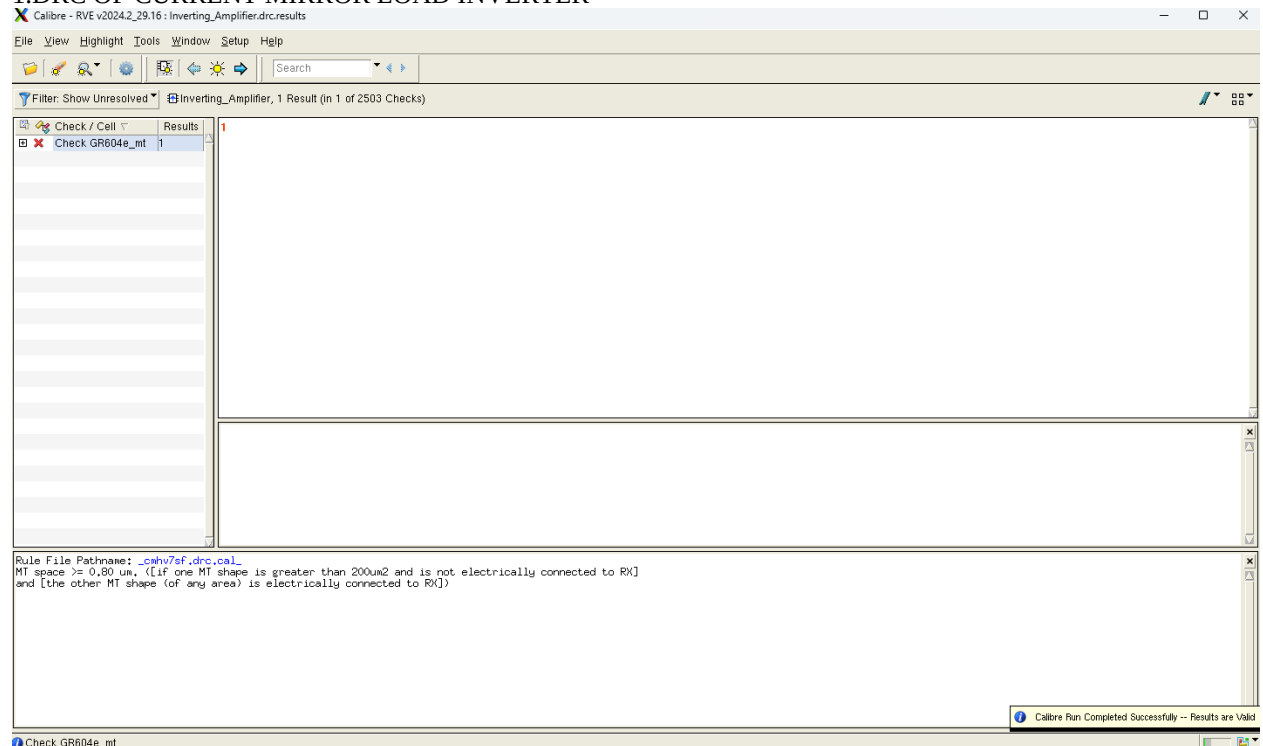


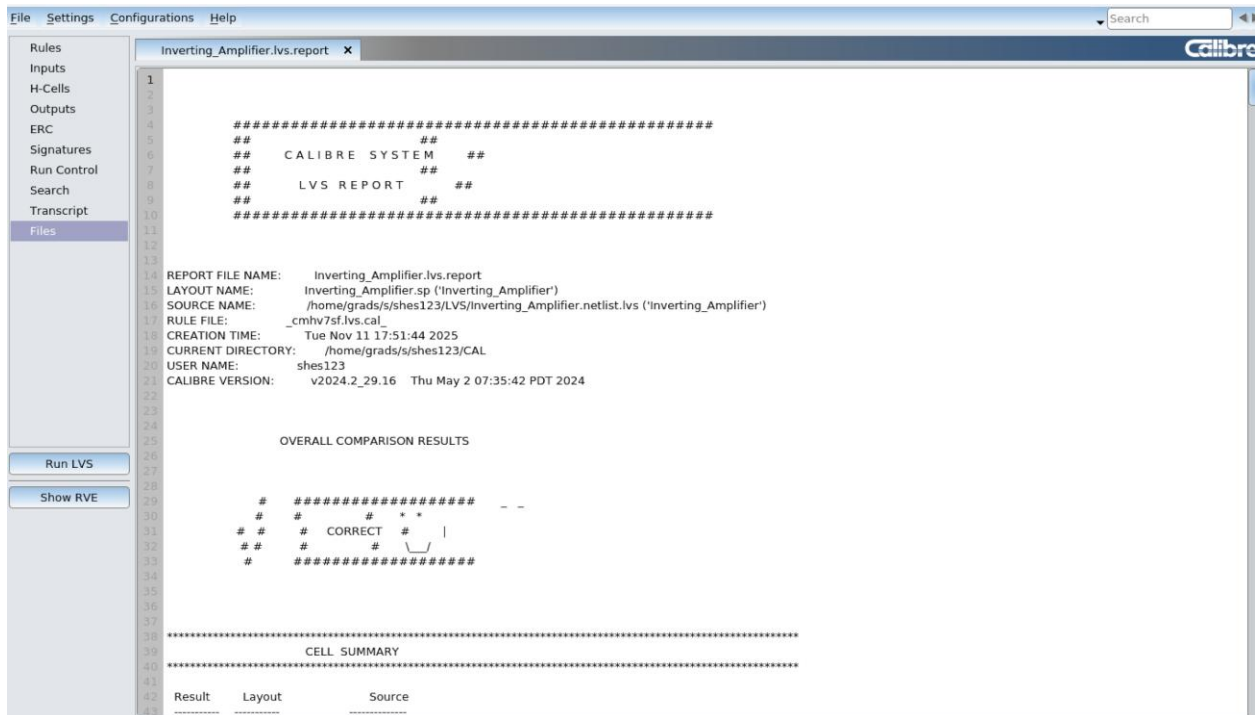
V. DRC AND LVS RESULTS

This section should include the DRC RVE window results for each layout cell (with the results filtered to “show unresolved” to show only the remaining DRC violations), and the cell name should be visible at the top of the DRC RVE window. The LVS report summary image should also be included for each cell with your NetID, the time stamp, and the smiley face/check mark visible to show that LVS passed.

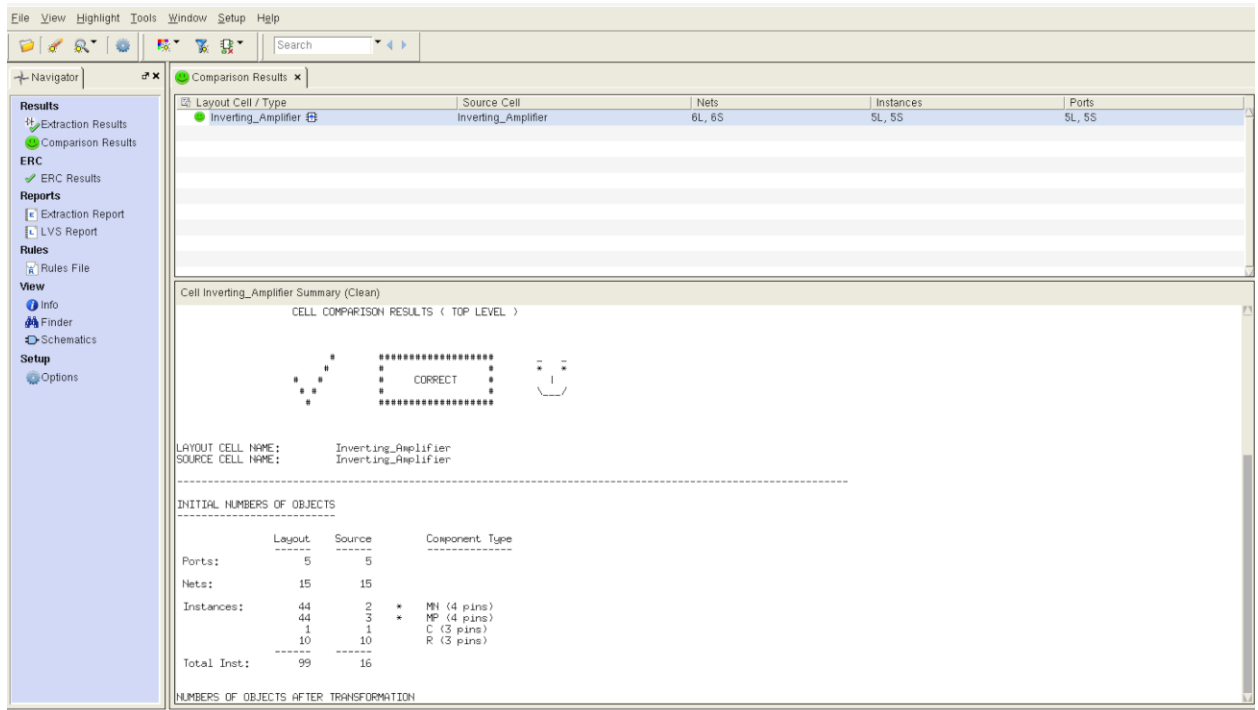
For the DRC errors, the first 2 inverters showed Antenna errors while the other two circuits had clean DRC and LVS

1. DRC OF CURRENT MIRROR LOAD INVERTER

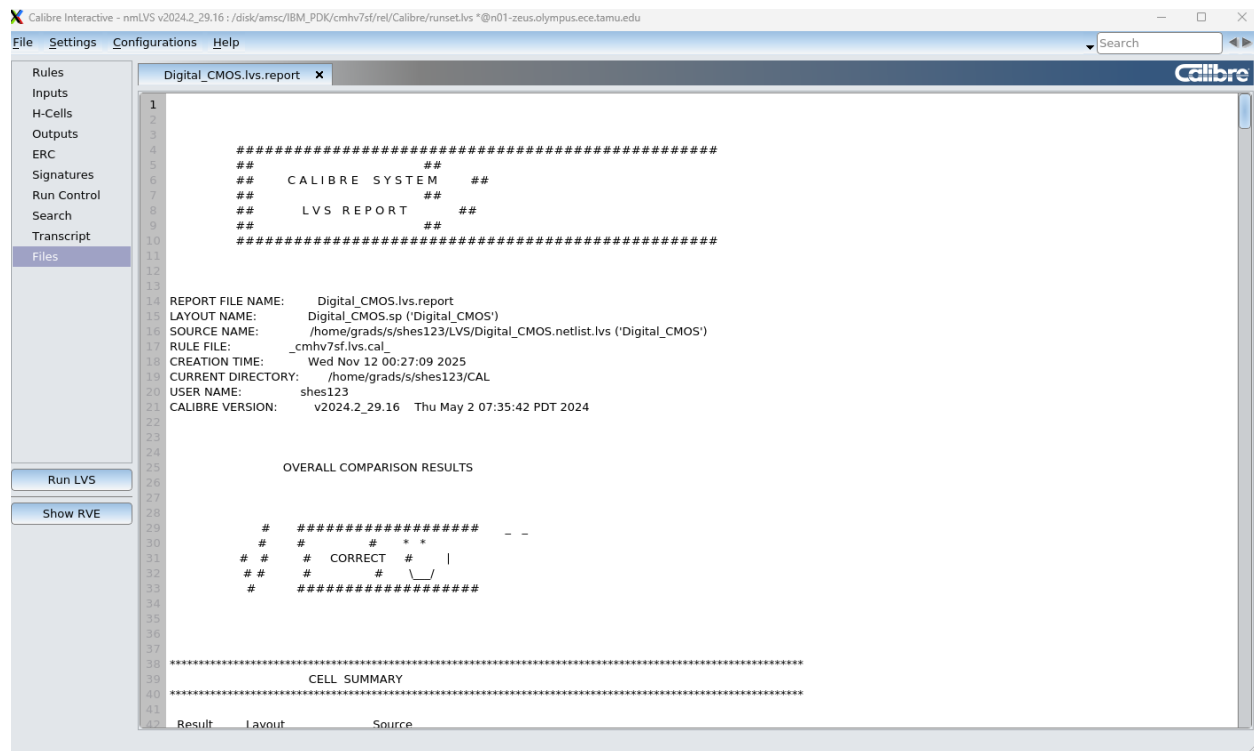




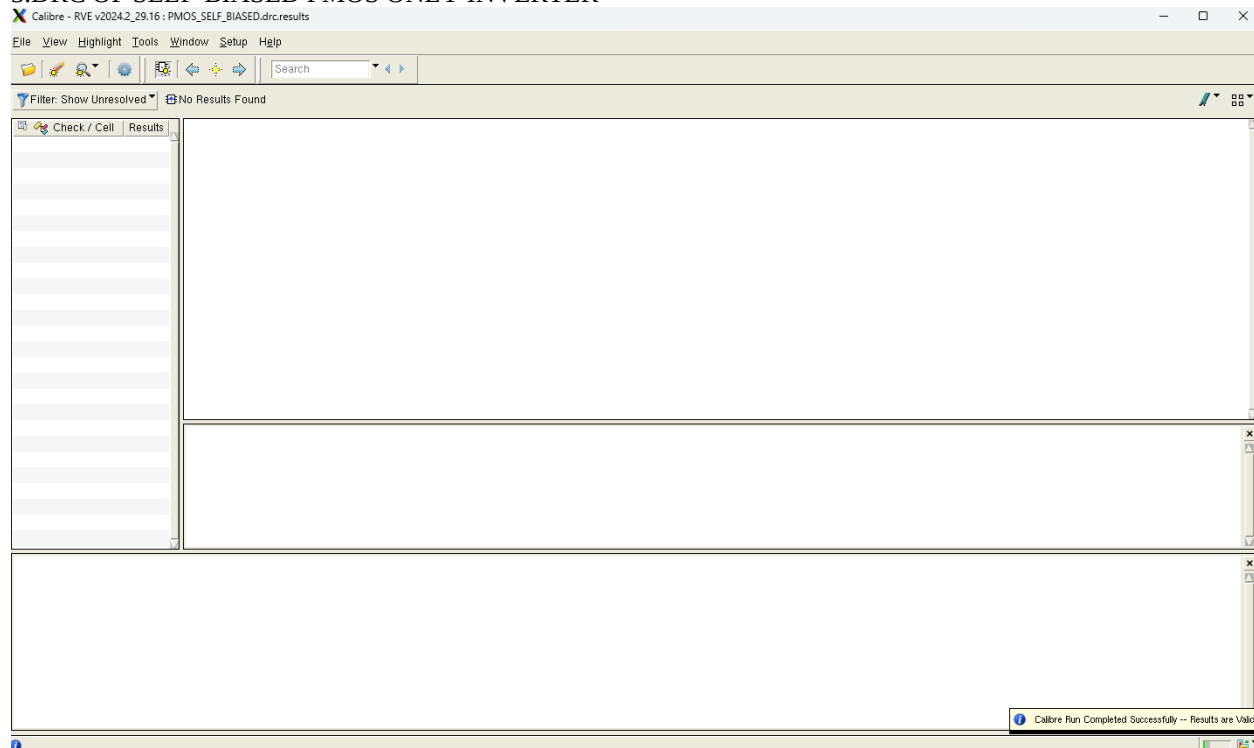
LVS OF CURRENT MIRROR LOAD INVERTER



2.DRC OF DIGITAL CMOS INVERTER



3.DRC OF SELF-BIASED PMOS ONLY INVERTER



LVS OF SELF-BIASED PMOS ONLY INVERTER

Calibre - RVE v2024.2_29.16 : svdb PMOS_SELF_BIASED

File View Highlight Tools Window Setup Help

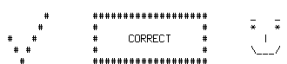
Search

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
PMOS_SELF_BIASED	PMOS_SELF_BIASED	5L, 5S	4L, 4S	4L, 4S

Cell PMOS_SELF_BIASED Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)



LAYOUT CELL NAME: PMOS_SELF_BIASED
SOURCE CELL NAME: PMOS_SELF_BIASED

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component	Type
Ports:	4	4		
Nets:	14	14		
Instances:	70	3	*	MP (4 pins)
	1	1		C (3 pins)
	10	10		R (3 pins)
Total Inst:	81	14		

NUMBERS OF OBJECTS AFTER TRANSFORMATION

Calibre Interactive - nmlVS v2024.2_29.16 : /disk/amsc/IBM_PDK/cmhv7sf/rel/Calibre/runset.lvs *@n01-zeus.olympus.ece.tamu.edu

File Settings Configurations Help

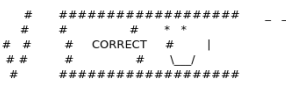
Search

Rules Inputs H-Cells Outputs ERC Signatures Run Control Search Transcript Files

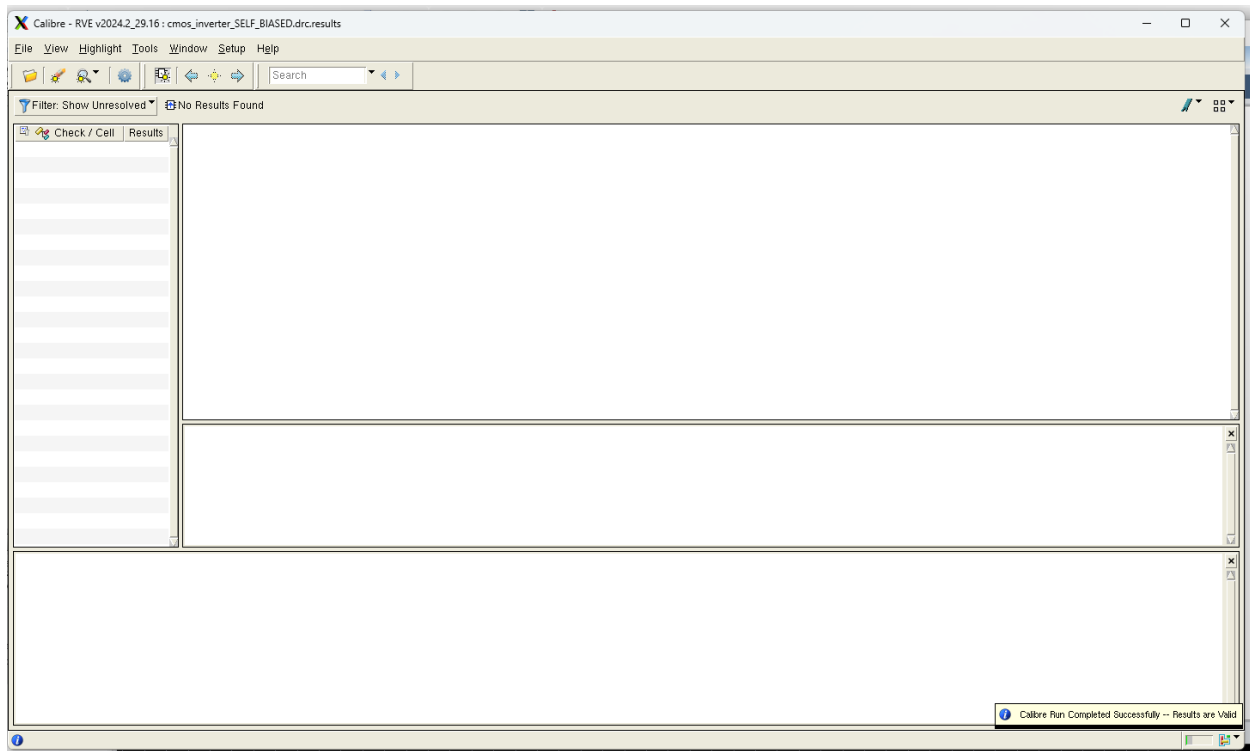
PMOS_SELF_BIASED.lvs.report

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41
42#####
CALIBRE SYSTEM ##
LVS REPORT ##
#####REPORT FILE NAME: PMOS_SELF_BIASED.lvs.report
LAYOUT NAME: PMOS_SELF_BIASED.sp ('PMOS_SELF_BIASED')
SOURCE NAME: /home/grads/s/shes123/LVS/PMOS_SELF_BIASED.netlist.lvs ('PMOS_SELF_BIASED')
RULE FILE: _cmhv7sf.lvs.cal_
CREATION TIME: Wed Nov 12 04:23:09 2025
CURRENT DIRECTORY: /home/grads/s/shes123/CAL
USER NAME: shes123
CALIBRE VERSION: v2024.2_29.16 Thu May 2 07:35:42 PDT 2024

OVERALL COMPARISON RESULTS

*****
CELL SUMMARY

4.DRC OF SELF-BIASED CMOS INVERTER



LVS SELF-BIASED INVERTER

Calibre - RVE v2024.2_29.16 : svdb cmos_inverter_SELF_BIASED

File View Highlight Tools Window Setup Help

Navigator Comparison Results x

Results

- Extraction Results
- Comparison Results

ERC

- ERC Results

Reports

- Extraction Report
- LVS Report

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

- Options

Layout Cell / Type	Source Cell	Nets	Instances	Ports
cmos_inverter_SELF_BIASED	cmos_inverter_SELF_BIASED	5L, 5S	4L, 4S	4L, 4S

Cell cmos_inverter_SELF_BIASED Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

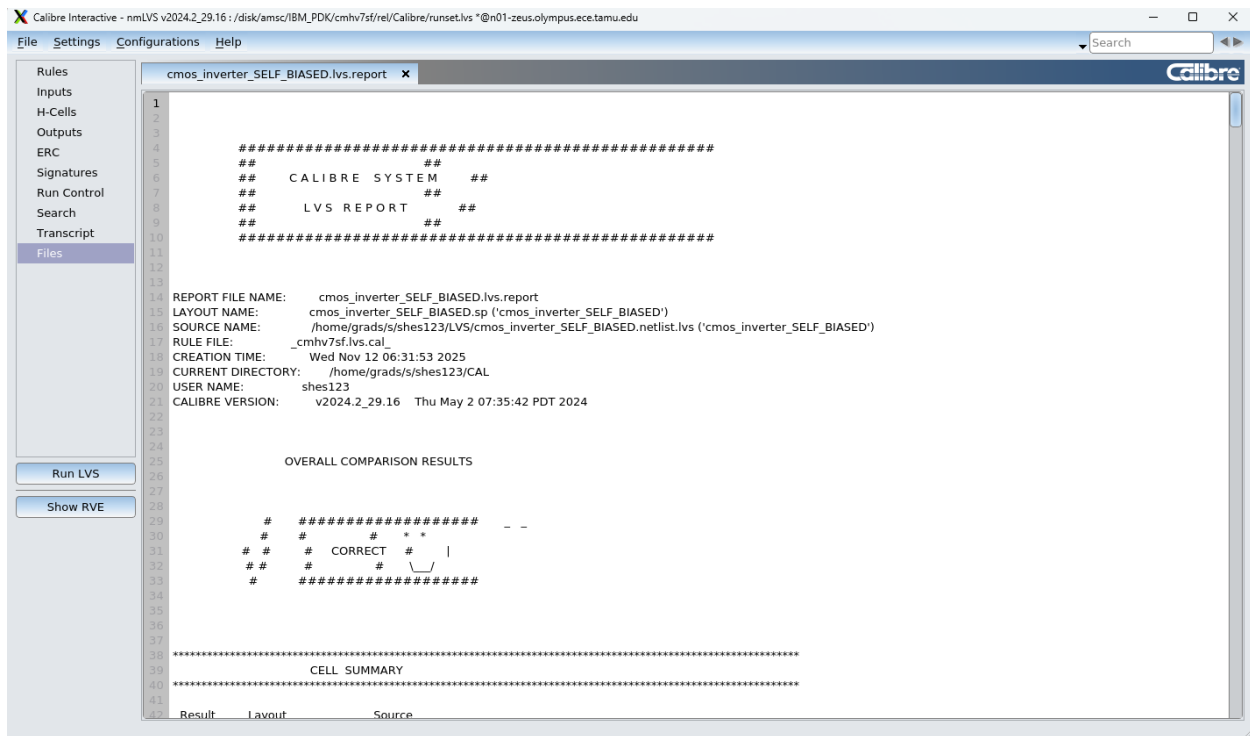
CORRECT

LAYOUT CELL NAME: cmos_inverter_SELF_BIASED
SOURCE CELL NAME: cmos_inverter_SELF_BIASED

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	4	4	
Nets:	14	14	
Instances:	18	2	* MN (4 pins)
	32	2	* MP (4 pins)
	1	1	C (3 pins)
	10	10	R (3 pins)
Total Inst:	61	15	

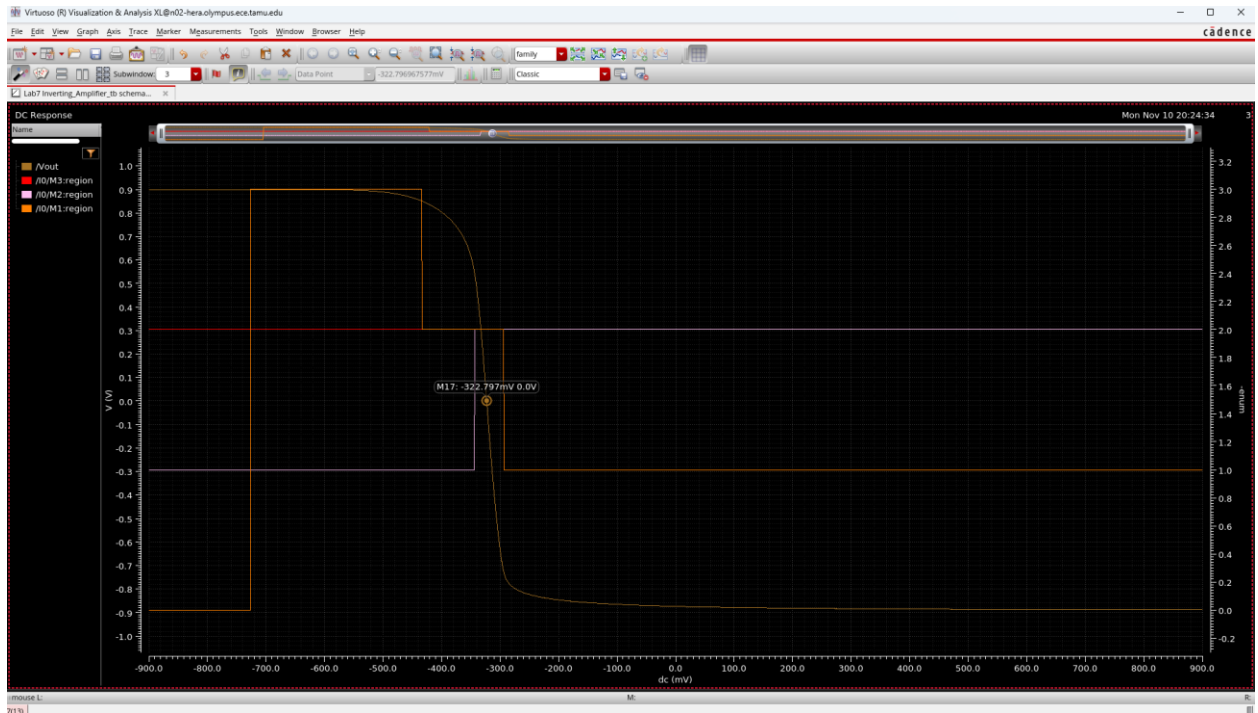
NUMBERS OF OBJECTS AFTER TRANSFORMATION



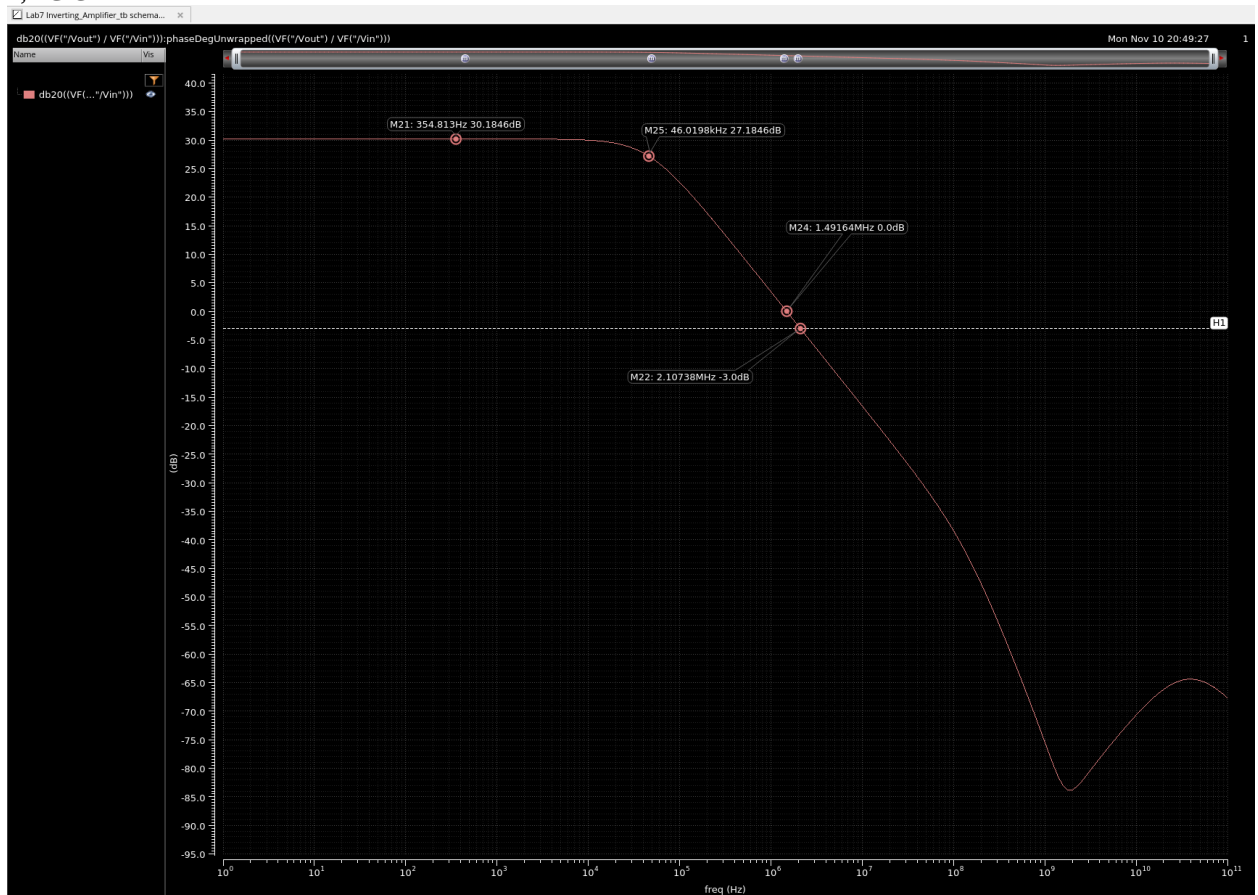
VI. SIMULATION RESULTS

Images of simulation results requested in the lab manual should be included here. Plots should be annotated to show relevant data points which will include either (1) data points that are explicitly requested in the lab manual or (2) data points from which other requested parameters are calculated. Data in the images should also be legible, so scale the images accordingly to make sure the data is easy to read. Post-layout simulations should be plotted in the same window with the pre-layout simulation results in order to make the differences between the two more apparent.

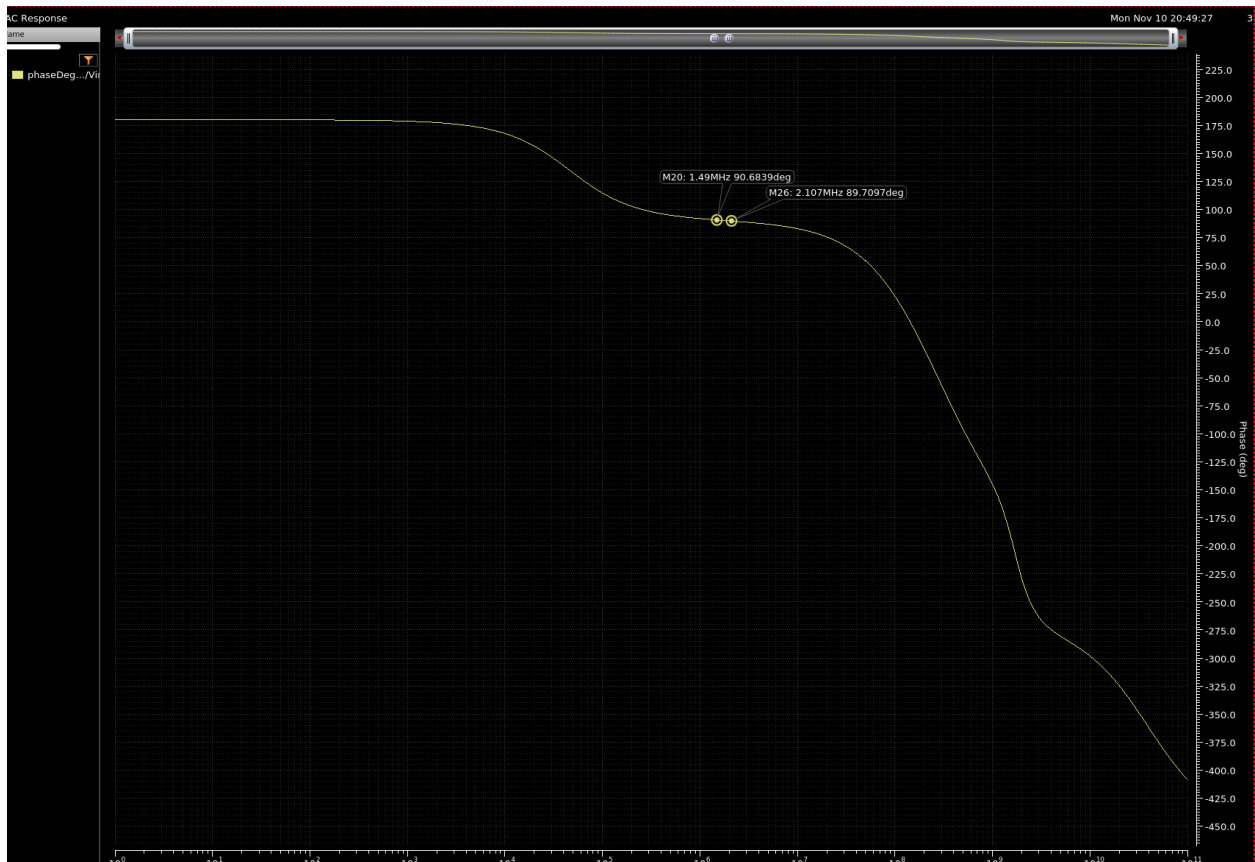
1.DC OPERATING POINT – CURRENT MIRROR LOAD INVERTER



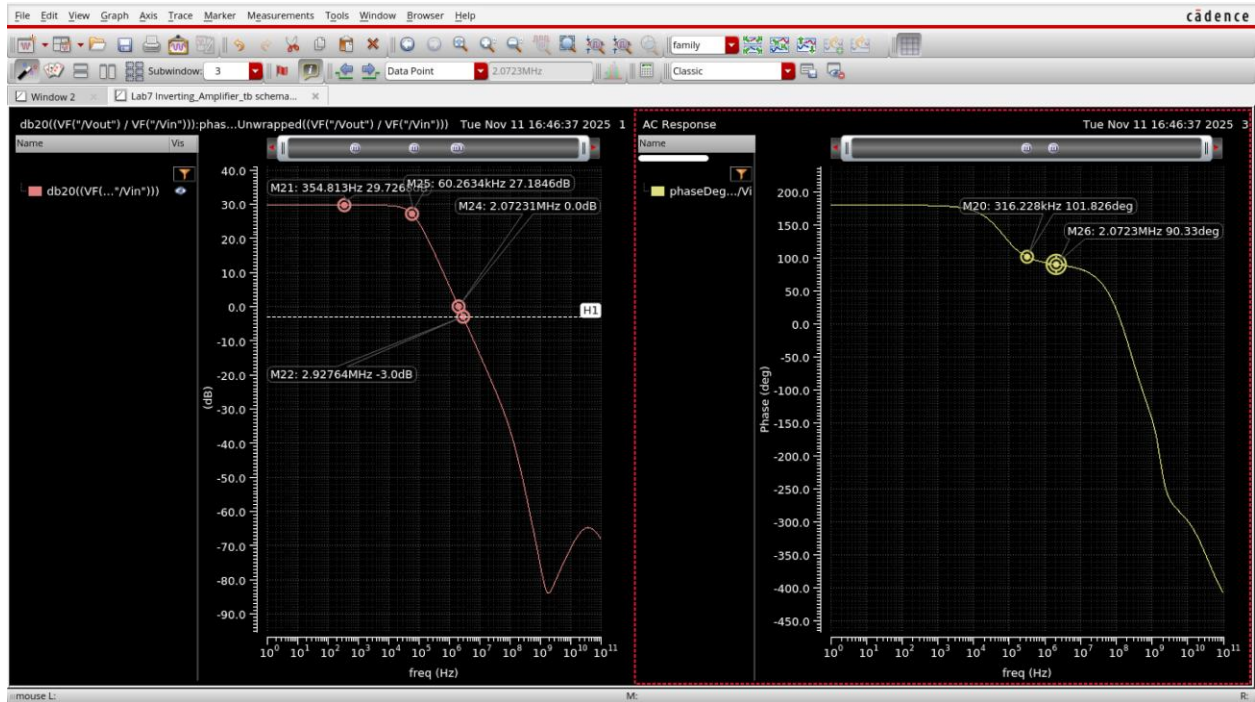
2, AC GAIN



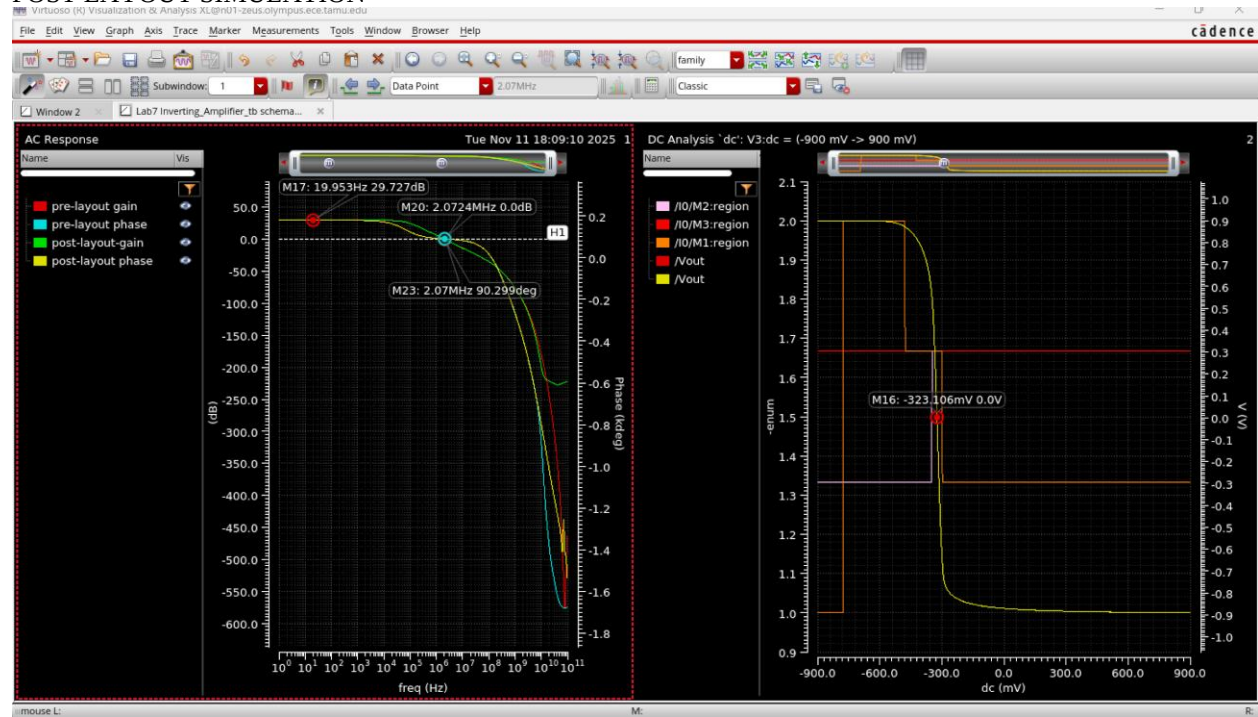
AC PHASE:



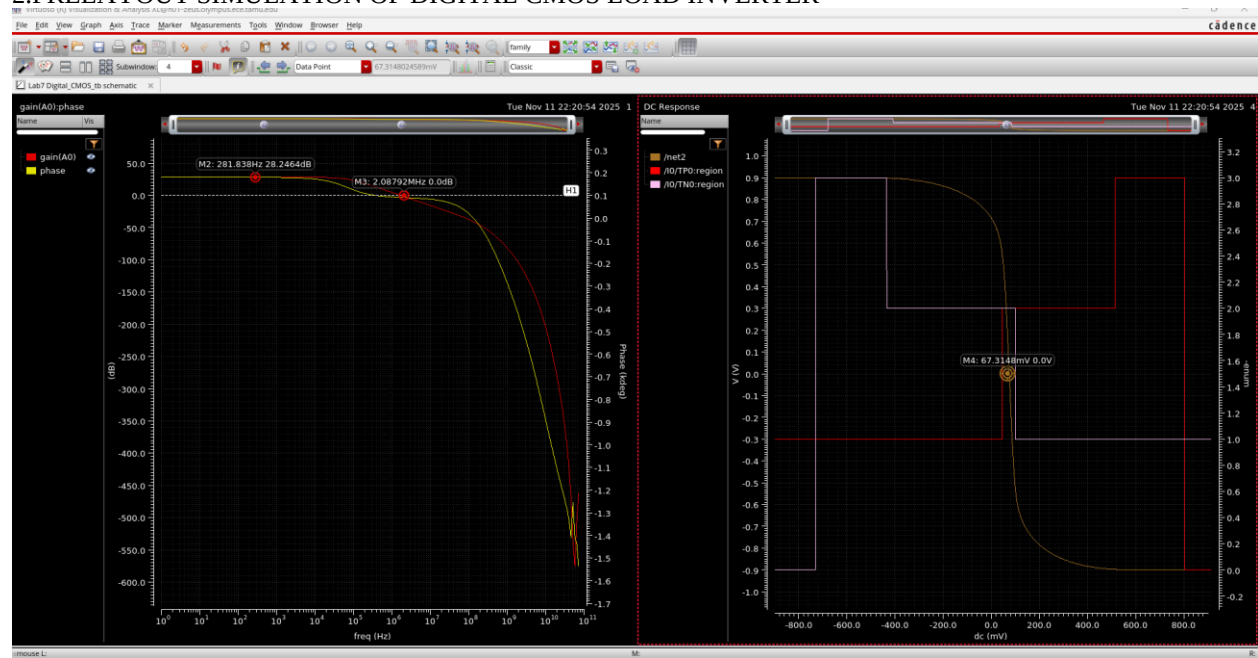
PRE-LAYOUT SIMULATION

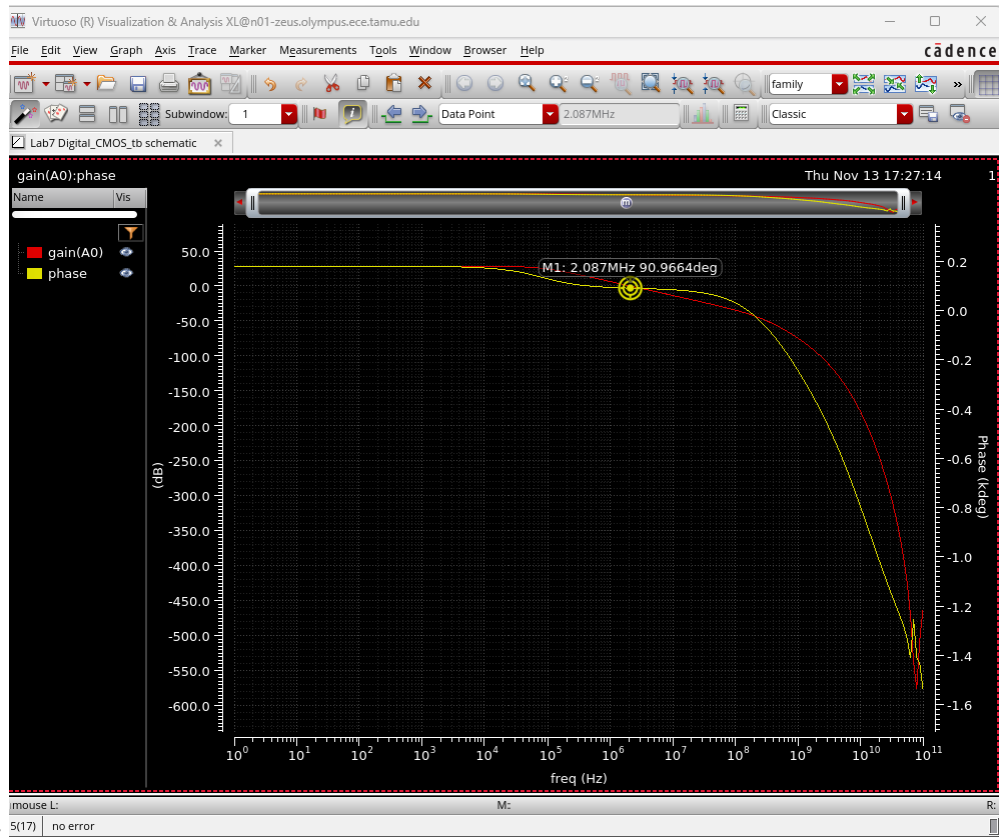


POST LAYOUT SIMULATION



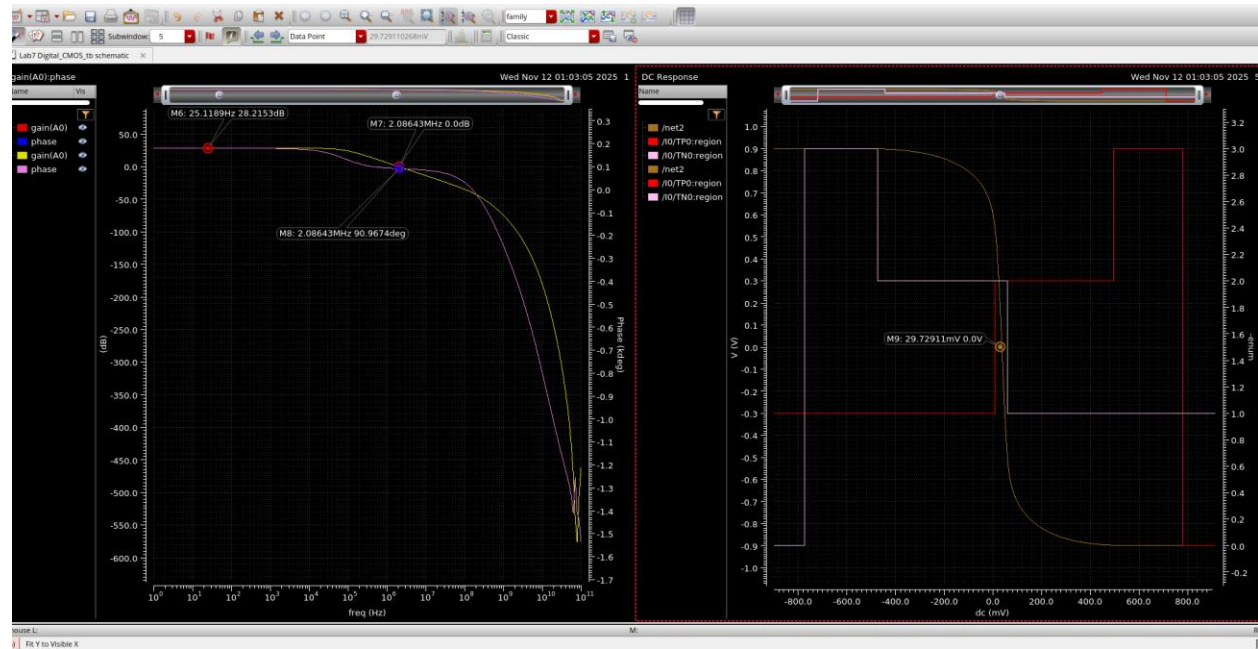
2. PRELAYOUT SIMULATION OF DIGITAL CMOS LOAD INVERTER



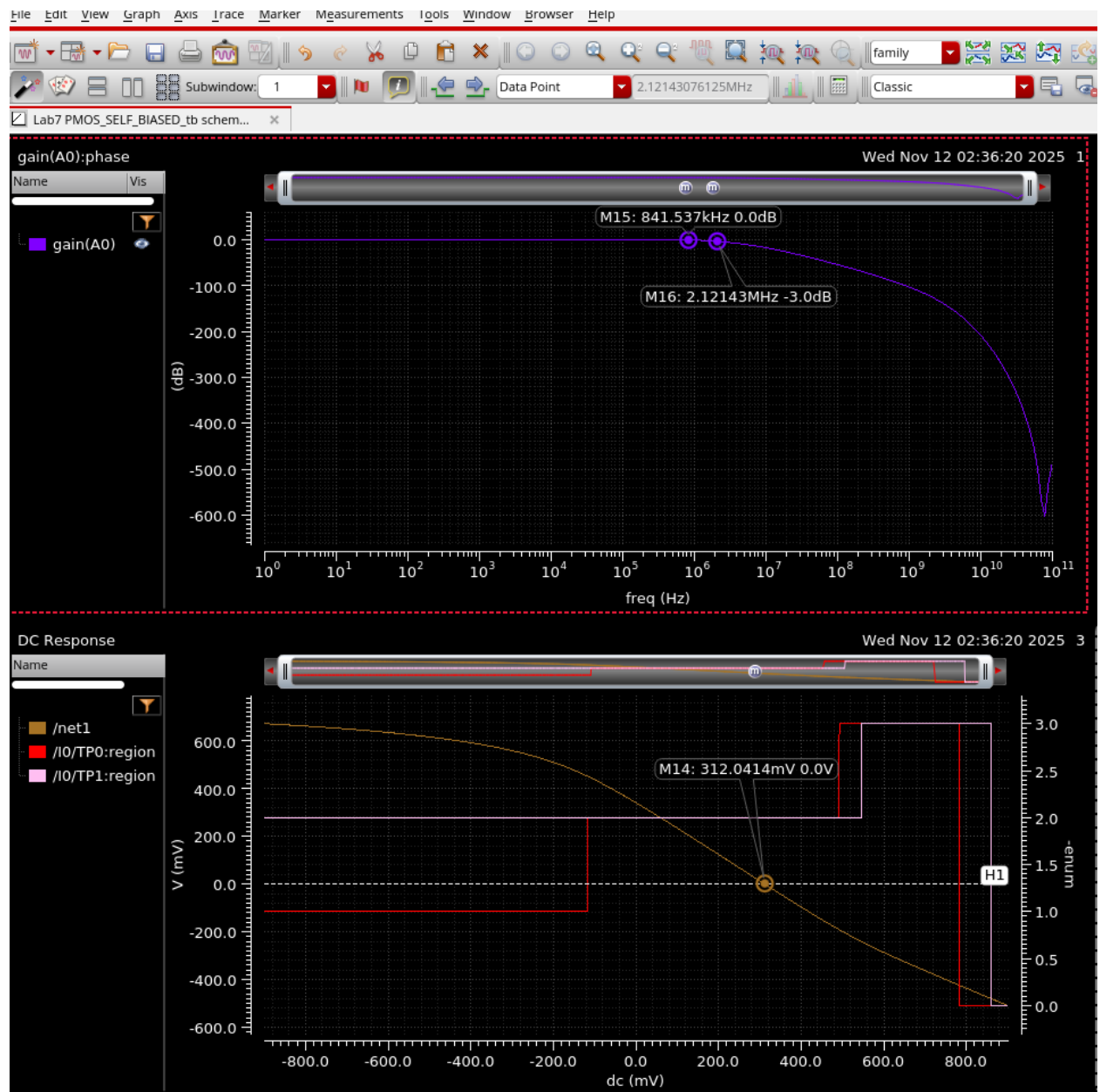


PHASE:

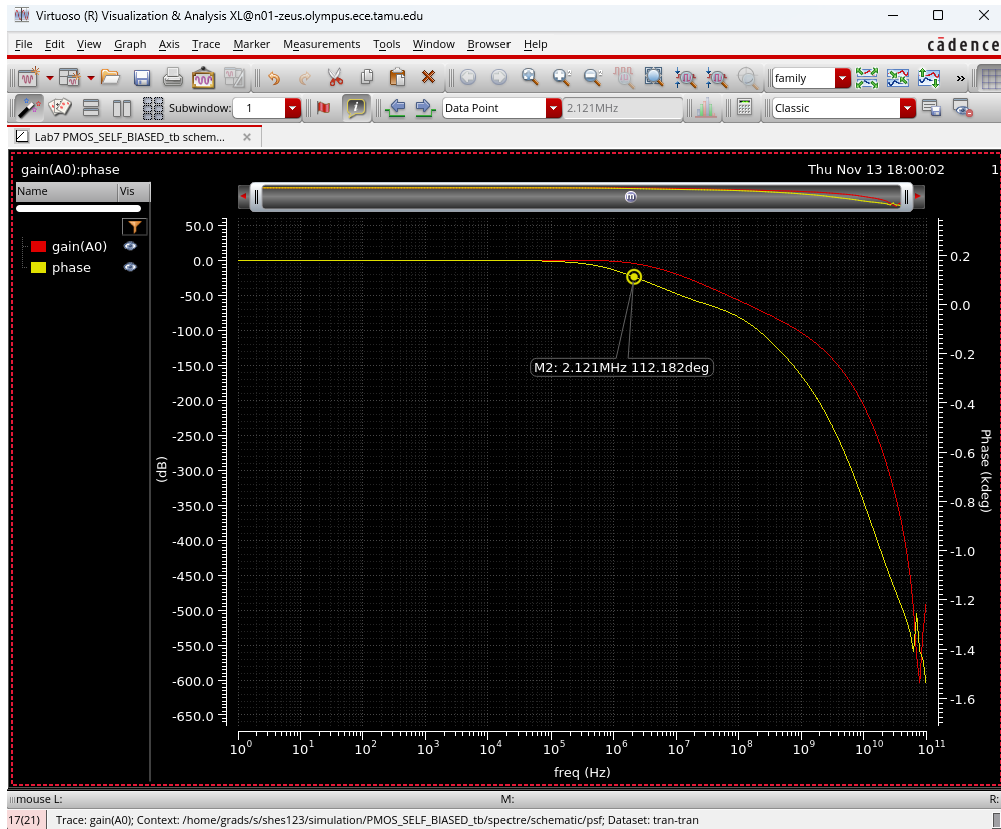
POST LAYOUT SIMULATION



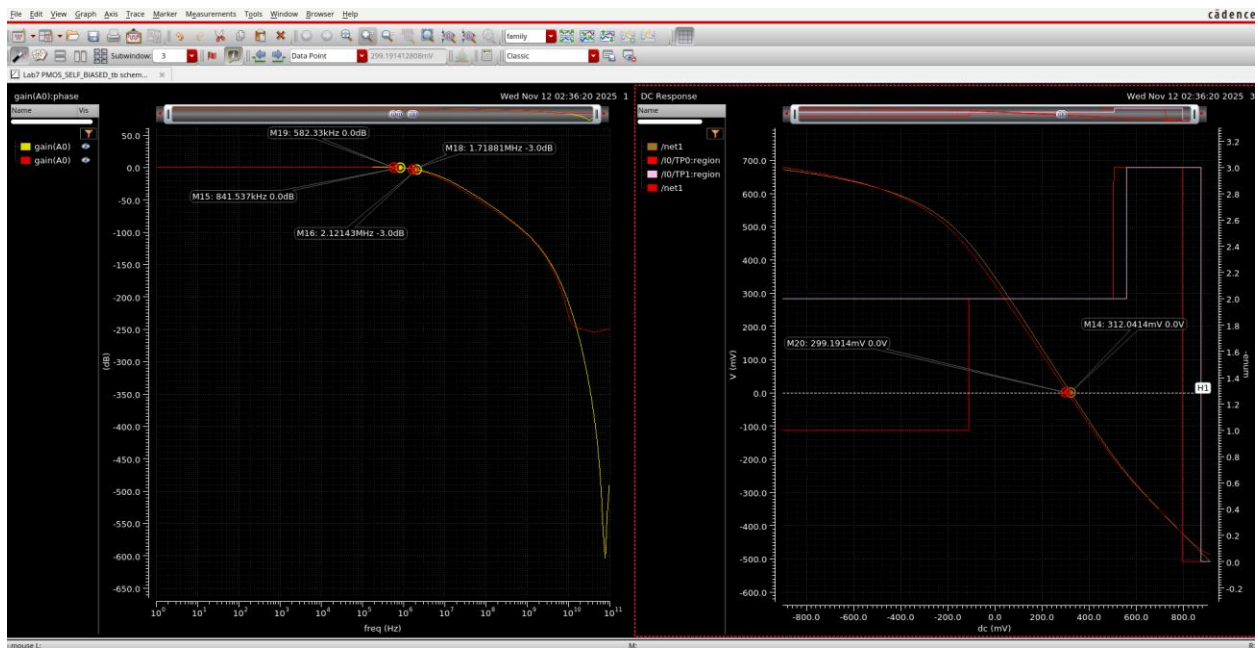
3.PRE-LAYOUT SIMULATION-SELF BIASED PMOS ONLY INVERTER



PHASE:

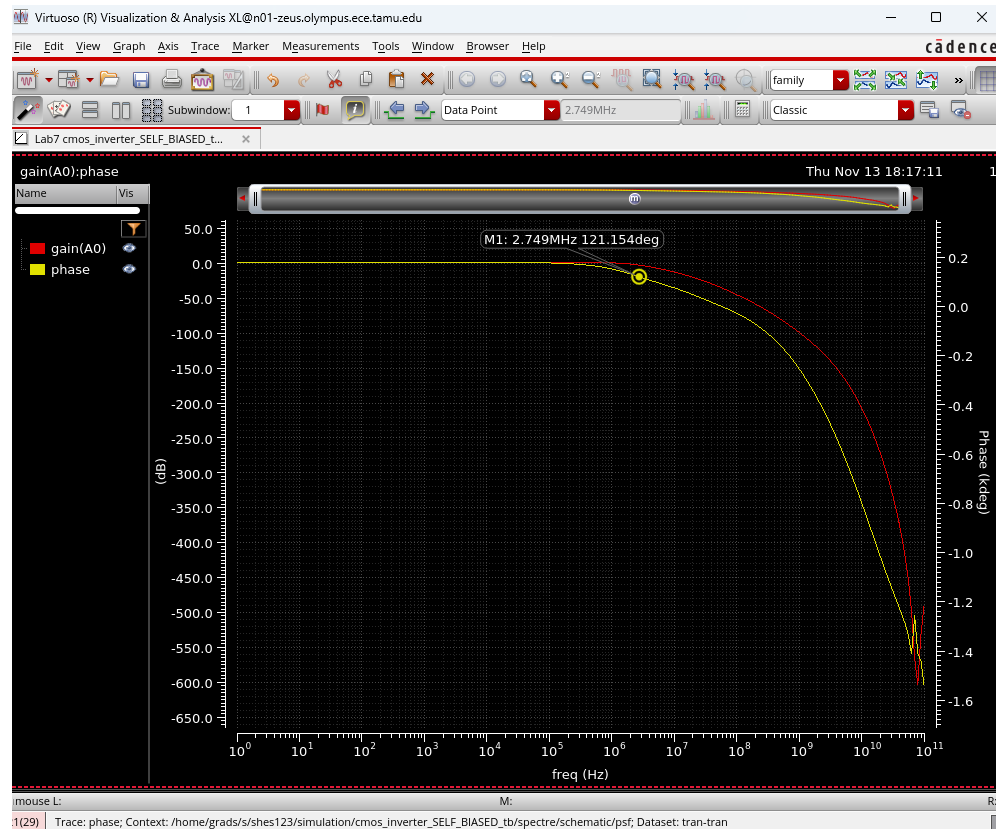


POST-LAYOUT SIMULATION

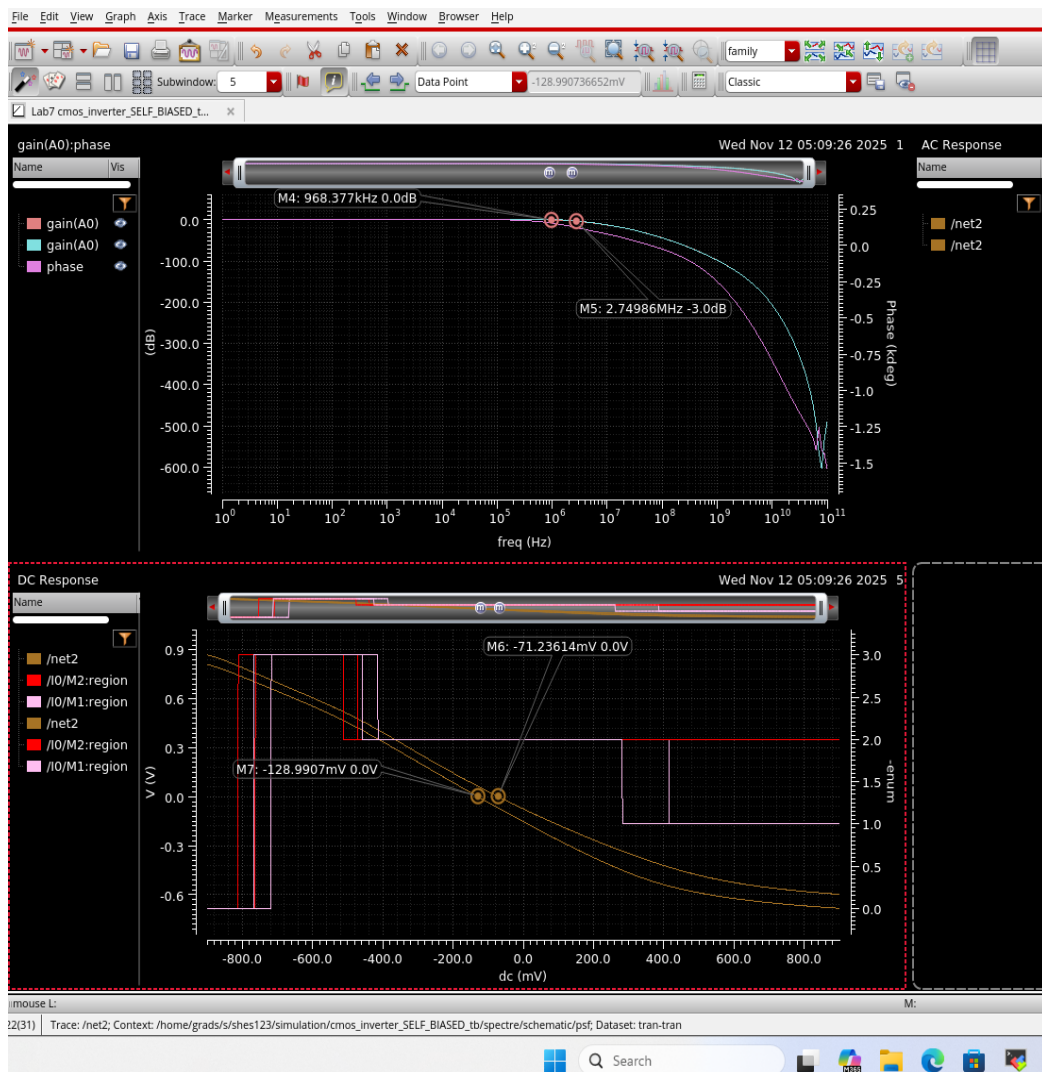




Phase:UGF:



5.POST-LAYOUT SIMULATION OF SELF-BIASED CMOS INVERTER





PHASE:UGF:

